Beyond Appearance: The Socioeconomic and Historical Roots of Racial Identity in Brazil

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Abstract

Using machine learning-based facial analysis and data on approximately 520,000 candidates for local office in Brazil, we study how self-reported racial identity relates to phenotypic appearance, socioeconomic characteristics, and local social context. A Shapley-Owen decomposition reveals that municipality fixed effects explain more variation in self-reported race than appearance, and their relative importance rises sharply among individuals with racially ambiguous phenotypes. Socioeconomic factors such as education exhibit "social whitening" effects concentrated among ambiguous individuals. Local social context has deep historical roots: areas with higher slave population shares in 1872 exhibit stronger norms favoring white self-identification, especially among the phenotypically ambiguous. A mediation analysis reveals that contemporary racial composition—but not economic channels—accounts for a substantial share of this association.

Keywords: Racial Classification, Social Identity, Phenotypic Appearance, Historical Legacy
JEL Codes: J15, N36, Z13

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1 Introduction

Racial inequality remains a pervasive feature of modern societies, shaping access to education, employment, income, political power, and justice (Derenoncourt, Kim, Kuhn and Schularick, 2024; Lang and Spitzer, 2020). Understanding the roots and persistence of these disparities requires attention not only to outcomes but also to how racial categories themselves are constructed and maintained. Current scholarship increasingly recognizes that racial classification is not only biologically meaningless (Morning, 2011) but also deeply embedded in social norms, cultural narratives, and historical legacies (Omi and Winant, 2014). How individuals are perceived racially—and how they identify themselves—reflects these layered social processes, making racial identity both fluid and context-dependent. This complexity is central to understanding how race contributes to persistent disparities in economic and social outcomes.

This paper investigates how self-reported racial identity in Brazil is shaped by physical appearance, socioeconomic characteristics, and social context. Brazil is marked by significant racial diversity, the result of European colonization and the slave labor of millions of Africans. This historical mix, along with the presence of Indigenous populations, has produced a wide range of phenotypic traits that do not always align neatly with broad racial categories such as white or black. Although the mixed-race category *pardo* (brown/mixed-race) is officially recognized in national statistics, racial identity in Brazil is widely understood to be a fluid and socially constructed phenomenon (Telles, 2004).

A key challenge in studying racial self-identification is obtaining comparable measures of individuals' physical appearance, which are essential for analyzing how people with similar phenotypic traits identify racially. To address this, we use a dataset of Brazilian candidates for local public office in the 2020 and 2024 municipal elections. As part of Brazil's electronic voting system, the Electoral Court collects and publicly releases candidate photographs that must meet specific quality standards. We apply machine learning techniques to these images to estimate, for each individual, the probability of being classified into one of six broad ethnoracial categories: *Asian*, black, *Indian*, *Latino/Hispanic*, *Middle Eastern*, and white. These probabilities serve as a multidi-

mensional measure of each candidate's appearance. Our main analysis uses the 2020 cross-section (approximately 520,000 candidates), and we validate our findings using a panel of candidates who ran in both elections, for whom we average appearance probabilities across the two photographs to reduce measurement error.

Candidates are also required to report a range of socioeconomic characteristics, including self-identified race, gender, age, marital status, occupation, and educational attainment. We use self-reported race as our main dependent variable, constructing a binary indicator equal to one if the candidate identifies as white and zero otherwise. The remaining variables serve as controls to explain variation in self-identification. Additionally, because we focus on local elections, each candidate can be directly linked to a specific municipality. We leverage this structure to include municipality fixed effects, which allow us to both control for and quantify local-level variation in social perceptions of race.

Our results suggest that, in a linear probability model, variation in phenotypic appearance explains approximately 23% of the variation in self-reported race. Compared to the reference category, *Latino/Hispanic*, individuals with a higher probability of being classified as white or *Middle Eastern* are more likely to self-identify as white, while those classified as black or *Indian* are less likely to do so. These patterns support the validity of our machine learning-based probabilities as a measure of phenotypic appearance. To decompose the relative importance of appearance, socioeconomic characteristics, and local context, we apply the Shapley-Owen method, which provides an order-invariant allocation of explained variation. The full model explains approximately 40% of the variation in self-reported race. Municipality fixed effects account for the largest share, followed by appearance probabilities, while socioeconomic characteristics contribute an order of magnitude less. Notably, appearance's contribution declines sharply with racial ambiguity: for individuals whose appearance does not clearly signal a racial category, municipality-level factors account for nearly 90% of explained variation, while appearance explains almost none.

With respect to socioeconomic characteristics, we find evidence consistent with "social whitening": conditional on phenotypic appearance, individuals with more favorable socioeconomic pro-

files are more likely to identify as white. Individuals with tertiary education are approximately 16 percentage points more likely to self-identify as white than those with less than primary education; adding appearance controls attenuates this to approximately 8 percentage points, and further adding municipality fixed effects leaves it at 8.1 percentage points. The roughly 50% attenuation reflects the correlation between education and phenotype in Brazil's historically stratified society, but the large residual effect is consistent with socioeconomic status shaping racial self-identification beyond what appearance can explain. Similar patterns hold for age. The male coefficient is negative, indicating that men are less likely than women to self-identify as white; however, after controlling for appearance and municipality fixed effects, the effect is only one percentage point, suggesting that the gender has a much smaller whitening effect than education and age.

To assess whether social whitening operates more strongly for individuals whose appearance is harder to classify, we examine how socioeconomic effects vary with racial ambiguity. We measure ambiguity using a logit-based index that is highest when the projected probability of white self-identification is close to 50%—indicating maximum phenotypic uncertainty—and lowest when the classification is unambiguous. The results reveal that social whitening is concentrated among racially ambiguous individuals: the effect of tertiary education on white self-identification approximately doubles from roughly 5–6 percentage points for those with unambiguous appearance to 9–10 percentage points for those with highly ambiguous appearance. The wealth index—a measure of predicted income constructed from observable demographics—also shows a positive gradient with ambiguity, paralleling the patterns for education and age. These findings indicate that when phenotypic cues provide a clear racial signal, socioeconomic characteristics matter less for self-identification; but when appearance is ambiguous, education, age, and wealth become substantially more important.

Finally, we investigate the institutional and historical roots of local social context. We estimate municipality fixed effects from a logistic regression controlling for appearance and socioeconomic characteristics, and interpret these as capturing local social context—the extent to which individuals with similar phenotypes and backgrounds are more or less likely to identify as white depending

on where they live. This context is significantly stronger in municipalities in the Southeast and South of Brazil, regions marked by both greater European immigration and a more intense slavery shares in the late nineteenth century.[1] We find that the share of enslaved individuals in 1872 is positively associated with white self-identification, even after controlling for state fixed effects, geographic characteristics, and contemporary demographics. This effect is concentrated among racially ambiguous individuals: for those whose appearance clearly signals white or nonwhite, historical slavery has little effect on self-identification, but for those with ambiguous phenotypes, the effect is large and highly significant. A mediation analysis reveals that local racial composition—the contemporary share of white residents—accounts for the majority of the slavery association, while a substantial direct effect persists even after conditioning on racial composition, consistent with an independent normative channel through which historically high-slavery areas fostered persistent social pressure to identify as white.

Related Literature. A large body of work establishes that racial identity is fluid and context-dependent. Individuals shift their self-identification in response to socioeconomic mobility (Saperstein and Penner, 2012; Schwartzman, 2007), social aspirations (Mitchell, 2010), anti-immigrant sentiment (Antman and Duncan, 2024), and political realignment (Agadjanian and Lacy, 2021); Saperstein (2025) provides a comprehensive review. In Latin America, the mapping from skin color to identity categories varies by national context, with status leading to whitening, darkening, or polarization depending on the country (Telles and Paschel, 2014; Bailey and Telles, 2006). In Brazil specifically, racial identification has been "darkening" since 2010 as affirmative action matures (Freeman, Telles and Goldberg, 2025) and peer networks shape how students identify (Portella, Kirschbaum and Menezes-Filho, 2022). There is also evidence that local racial composition modifies how ancestry maps to self-declared race (Chor, Pereira, Pacheco, Santos, Fonseca, Schmidt, Duncan, Barreto, Aquino, Mill, Molina, Giatti, Almeida, Bensenor and Lotufo, 2019) and how politicians change their racial identity (Janusz, 2021). We add to this literature by going

[1] This is especially true for the Southeast region, where Rio de Janeiro had about one-quarter of population composed by slaves.

beyond the question of whether local context affects racial self-identification to quantify its relative importance compared to socioeconomic characteristics and phenotypic appearance. Moreover, we provide evidence that norms shaped by historical slavery may drive local context effects, and that this operates through cultural transmission rather than economic channels.

Theory predicts that the social construction of race should operate most strongly at the boundary of classification. McGee (2025) formalizes this intuition: elites use observable, heritable color-based categories to maintain hierarchies, creating distinctions between arbitrarily similar people on opposite sides of the color line. This paper also provides theoretical foundations for the positive effect of slavery on white self-identification. The model implies that in localities with larger shares of slaves, elites would be more willing to grant "white status" to mixed-race individuals to avoid social unrest induced by the coalition between them and black slaves. Our findings suggest that these norms of classification persist to the present day, and that they are particularly salient for individuals whose phenotypic appearance is ambiguous.

Empirically, Francis and Tannuri-Pianto (2013) confirm this boundary logic in Brazil: socioeconomic status effects on racial self-identification are concentrated among individuals near the transitions between skin tone quintiles, and affirmative action policies further reshape identification patterns (Francis-Tan and Tannuri-Pianto, 2024; Antman and Duncan, 2023). We extend this insight by constructing a continuous measure of racial ambiguity and showing that the relative importance of social and contextual factors rises monotonically as phenotypic signals become less definitive—a pattern that Francis and Tannuri-Pianto (2013) document at discrete skin tone boundaries but that no prior study has traced across the full ambiguity distribution.

Measuring phenotypic appearance at this level of resolution requires moving beyond existing approaches. Prior work relies on human-rated skin tone (Francis and Tannuri-Pianto, 2013), observer assessments of photographs (Feliciano, 2016), or genotype measure of ancestry (Chor et al., 2019). We instead use the full distribution of machine learning classification probabilities as a continuous, multidimensional phenotypic similarity metric. This probability vector is what enables the continuous ambiguity measure that anchors our heterogeneity analysis. The approach

complements the colorism literature (Monk Jr., 2016; Dupree-Wilson, 2021), which shows that skin color operates as a gradient in shaping stratification outcomes, by providing a tool to study how self-identification responds to that gradient. Baerlocher and Schneider (2026) also leverage our machine learning approach to study the effect of the election of municipal councilsors that self identify as white but whose appearance is more likely perceived as nonwhite on the racial distribution of public employment.

Finally, we connect racial self-identification to the literature on the long-term effects of historical institutions. Existing work links slavery to persistent economic inequality (Laudares and Valencia Caicedo, 2023; Naritomi, Soares and Assunção, 2012) and cultural mistrust (Nunn and Wantchekon, 2011). However, no prior study links historical slavery intensity to individual racial *self-identification* in the contemporary period. We fill this gap and show that the mechanism is cultural rather than economic: racial composition—not wage gaps, educational attainment, land inequality, or municipal development—accounts for the slavery association, echoing Nunn and Wantchekon's (2011) finding that cultural norms are at least twice as important as institutional channels in transmitting the slave trade's effects. Beyond the theoretical foundation provided by McGee (2025), historical *whitening* policies in Brazil provide a concrete mechanism for our findings.[2] Specifically, the tendency to classify mixed-race individuals as white in early censuses to promote national progress (Dos Santos, 2002) created norms of classification that persist to the present day in areas where mixed-race individuals were more salient, likely locations with a stronger slave presence (Skidmore, 2009).

2 Historical and Sociopolitical Background

Brazil was colonized primarily by the Portuguese, although other European groups—such as Italians, Germans, French, and Dutch—also contributed to its settlement. Long before colonization, however, the territory was home to a diverse array of Indigenous peoples, whose cultures and

[2]Similar patterns of historical *whitening* are studied by Adukia, Hornbeck, Keniston and Lualdi (2025) and Dahis, Nix and Qian (2020).

populations were profoundly impacted by European conquest, disease, and displacement. Over more than three centuries of colonial rule, Brazil also became a major destination in the transatlantic slave trade, with millions of Africans forcibly brought to work on sugarcane plantations and in other forms of coerced labor. The interaction among Indigenous, African, and European populations led to widespread racial mixing, which remains a defining feature of Brazilian society.

Given these diverse racial and ethnic origins that shaped Brazilian society, some researchers have argued that race is a meaningless or nonexistent category in Brazil—a notion often referred to as the "racial democracy" myth. However, this view is contradicted by a wide body of empirical evidence showing persistent racial inequalities (Andrews, 1996; Htun, 2004; Dupree-Wilson, 2021). Data from the Brazilian Institute of Geography and Statistics (IBGE) indicate that Afro-descendants in Brazil continue to face structural disadvantages: they have lower average incomes, higher unemployment rates, lower educational attainment, and are disproportionately represented in informal labor markets (IBGE, 2024). They are also more likely to live in underserved areas with limited access to quality healthcare, education, and public services. These disparities highlight how race continues to operate as a powerful axis of social and economic inequality in contemporary Brazil.

The fluidity of racial categories in Brazil was not simply an organic outcome of racial mixing—it was actively constructed by the state. After the abolition of slavery in 1888, the government pursued an explicit *branqueamento* (whitening) policy, subsidizing mass European immigration as a strategy for transforming the country's racial composition and linking national progress to the gradual dilution of Afro-descendant and Indigenous populations (Dos Santos, 2002). This ideology was inscribed directly into official statistics: the 1890 Census removed the *pardo* category and reclassified mixed-race individuals as white, deliberately inflating the reported white population share, while the 1920 Census omitted the race question entirely (Camargo, 2009). When the census reintroduced racial categories in 1940, it again excluded *pardo*, replacing race with a "color" question; the category was dropped once more in 1970 and fully reinstated only in 1991 (Camargo, 2009). These repeated definitional shifts reflect how racial classification in Brazil has always

been entangled with state objectives rather than social consensus. The historical valorization of whiteness left a lasting imprint on self-identification: mixed-race Brazilians internalized a tendency to adopt a white identity when social circumstances permitted. This tendency began to reverse from the 1990s onward, as Afro-Brazilian social movements and affirmative action policies encouraged identification with Afro-descendant categories; the 2022 Census, for the first time, reported a majority of respondents identifying as *pardo* (IBGE, 2024).

The Brazilian Census currently uses five racial categories—*branco* (white), *preto* (black), *pardo* (brown/mixed-race), *amarelo* (yellow/Asian), and *indígena* (Indigenous)—and classification is based solely on self-identification. One of the primary challenges in studying racial inequalities with self-classified data is the fluid and socially constructed nature of racial identity. In Brazil, where racial boundaries are often ambiguous and heavily influenced by social, regional, and economic factors, individuals may self-identify in ways that do not align with how others perceive them. This subjective element can obscure patterns of discrimination that are based on phenotypic traits such as skin color, facial features, or hair texture—characteristics that may drive discriminatory behavior regardless of how the target self-identifies.

Moreover, self-identification is shaped by socioeconomic mobility. The *branqueamento* ideology —which explicitly linked whiteness to social progress—created lasting incentives for individuals to claim a white identity as their social standing improved, a process known as "social whitening" (Schwartzman, 2007; Telles and Paschel, 2014). Critically, as we document in Section 6, this tendency is not uniform across space: it is stronger in areas with a greater historical slave presence, suggesting that the social valorization of whiteness engendered by slavery persists in contemporary self-classification. These patterns complicate efforts to track structural racial inequalities through self-reported data alone. If phenotypically nonwhite individuals self-classify as white yet continue to receive lower incomes due to their appearance, they are counted in the "white" group, mechanically compressing measured racial wage gaps. The true disparity between phenotypically white and phenotypically nonwhite individuals is therefore larger than the gap between self-reported white and self-reported nonwhite, and measured racial gaps may narrow partly

due to identity reclassification rather than a genuine reduction in discrimination.

The recent trend of self-declaration as nonwhite can be partially attributed to a series of affirmative action policies implemented by the Brazilian government creating benefits for nonwhite individuals in various contexts, for example, access to tertiary education (Vieira and Arends-Kuenning, 2019; Mello, 2022). In our setting, it is important to point out that there was no effective policy that incentivized individual candidates to self-declare as nonwhite. Although racial affirmative action in campaign finance nominally began in 2020—when the Supreme Federal Court ruled in ADPF 738 that public campaign funds and free advertising time must be distributed proportionally to the share of black and *pardo* candidates nominated by each party—the rule was largely ignored by parties requiring the congress to later approve Constitutional Amendment 133, which replaced it with a fixed floor of 30% of public campaign funds for black and *pardo* candidates and granted amnesty for past violations. Importantly, the rule applied to the aggregate allocation of funds across racial groups within a party, not to equal funding for each candidate: a party with 40% nonwhite candidates must allocate 40% of its funds to nonwhite candidates as a group, but can concentrate those resources on its most viable nonwhite candidate. Individual candidates therefore do not benefit from self-declaring as nonwhite, and their racial declarations likely reflected their own notions of race rather than strategic positioning.

3 Empirical Framework

3.1 Conceptual issues for racial self-identification

We hypothesize that an individual's self-reported race depends on three sets of variables: the phenotypic appearance of individual i, denoted by the vector $\mathbf{z}_i$; their socioeconomic characteristics, denoted by the vector $\mathbf{x}_i$; and the influence of their community, denoted by w_i. Let the reported race be represented by the binary variable y_i, indicating whether individual i reported being white. We model this as

$$y_i = f(\mathbf{z}_i, \mathbf{x}_i, w_i) + \varepsilon_i, \tag{1}$$

where ε_i is an error term assumed to have mean zero.

Appearance ($\mathbf{z}_i$) shapes self-identification because racial categories in Brazil, while socially constructed, remain anchored to phenotypic traits perceived by others (Telles and Paschel, 2014). Socioeconomic characteristics ($\mathbf{x}_i$) matter because individuals may shift their racial self-identification in response to social mobility—identifying with the group whose status matches their own (Schwartzman, 2007; Akerlof and Kranton, 2010). Community context (w_i) shapes identity through the local prevalence of racial categories, social norms about classification, and the reference groups individuals observe in their daily environments (Telles and Paschel, 2014; Bernheim, 1994).

The relative weight of these three components is an empirical question, and we expect it to vary with the ambiguity of an individual's appearance: when phenotype provides a clear racial signal, $\mathbf{z}_i$ should dominate; when appearance is ambiguous, social context and socioeconomic position should play a larger role. To see why, consider the informational role of the phenotypic signal. Let $\widehat{p}_i^{\text{white}}$ denote the logit-based probability that individual i self-identifies as white implied by the full appearance vector (see Section 3.4 for precise definition). When $\widehat{p}_i^{\text{white}} \approx 0$ or ≈ 1, appearance provides a strong, consistent signal about racial category, anchoring self-identification. When $\widehat{p}_i^{\text{white}} \approx 0.5$, the signal is weak, leaving the individual with more discretion over self-identification. Formally, the precision of the phenotypic signal is proportional to its distance from 0.5—that is, to $1 - A_i$, where the ambiguity measure is $A_i = 1 - 2|\widehat{p}_i^{\text{white}} - 0.5|$. When signal precision is high ($A_i \approx 0$), appearance dominates; when precision is low ($A_i \approx 1$), community context and socioeconomic position fill the gap left by the weak phenotypic anchor. This yields the prediction that the contribution of $\mathbf{z}_i$ to explained variation in y_i declines monotonically with A_i, while the relative contributions of $\mathbf{x}_i$ and w_i rise.

Our goal is to measure the relative importance of each set of variables in shaping individuals' self-reported race. If race were purely a matter of physical appearance, then $\mathbf{x}_i$ and w_i would have no effect. Thus, identifying the role of phenotypic appearance is central to our analysis. In this context, we treat Brazilian municipalities as the relevant communities. Brazil's approximately 5,500 municipalities range from São Paulo (12 million residents) to towns of under 1,000; for

large cities, the municipality fixed effect averages over substantial within-city variation in racial norms across neighborhoods and social environments, which the fixed effect cannot capture. This likely introduces attenuation in estimates of local-context effects for urban candidates, making our estimates conservative for that subgroup. In our primary analysis, f is assumed to be linear and w_i is a municipality fixed effect, so Equation (1) becomes a linear probability model (LPM) estimated by ordinary least squares (OLS). Coefficients are therefore interpretable directly as marginal effects on the probability of self-identifying as white. Appendix B confirms that logit specifications yield qualitatively identical results.

3.2 Data and Sample Construction

Our sample includes the universe of all candidates who ran for mayoral and municipal council offices in the 2020 Brazilian municipal elections—a total of 526,965 individuals, after restricting to first-round participants in standard municipal elections and removing duplicate and ineligible registrations (total registered entries: 558,741). Face detection reduces this to approximately 521,000 candidates, which form the regression sample. We focus on candidates in local elections for several reasons. First, they provide a large and diverse sample, much broader than using candidates in state or federal contests. Second, local candidates—especially those running for municipal council positions—are less likely to be "professional" politicians, which enhances the external validity of our findings. Third, local elections allow for an accurate linkage between each candidate and their municipality, enabling us to capture community-level effects.

It is important to note that candidates are self-selected into the sample. They are more likely to be male, formally educated, and integrated into public life—and thus do not represent the full distribution of racial self-identification in Brazil. The patterns we document, particularly the magnitude of social whitening effects and the role of local context, may differ in the broader population. However, local candidates—especially those running for municipal council positions—are substantially more representative of the general population than candidates for state or federal office. For example, the share of candidates identifying as white (48%) is close to the national share

(43%), and the share of married candidates (52%) is close to the Census (51%). The share of male candidates is substantially different: 66% of candidates are male, much higher than the national share (48%). In terms of education, the share of candidates with primary (17%), secondary (42%), and tertiary (24%) education is close to the national share (15%, 35%, 17%).

Importantly, as discussed above, there is no effective individual-level incentive for candidates to misreport their race. Beyond the absence of policy incentives, the candidacy registration form is an administrative declaration to a state institution that voters are unlikely to consult; voters instead infer a candidate's race from television ads, campaign materials, and other visual interactions (see Baerlocher and Schneider, 2026, for further discussion on this point). Strategic racial misreporting in the registration context is therefore implausible.

A distinct, more subtle concern is that the electoral registration context may induce social-desirability effects whereby candidates select a racial category that projects a desired public image. If declarations are systematically more performative in high-slave-share areas due to stronger local norms about public racial presentation, the slavery association we document would capture compliance with local norms rather than internalized identity—which would remain substantively interesting, as it is the identity that candidates present in public life and that shapes how they interact with institutions. Our data cannot fully distinguish these interpretations; we proceed treating the declared race as the relevant object of study.

While we acknowledge the caveats listed above, a key advantage of using electoral candidates is the public availability of standardized photographs. Due to Brazil's electronic voting system, candidate photos are submitted by political parties when registering candidacies and must meet uniform formatting standards established by the Electoral Court.

3.3 Measuring Phenotypic Appearance

We measure phenotypic appearance using the DeepFace pipeline (Serengil and Ozpinar, 2024; Serengil, 2026), an automatic facial analysis system that classifies individuals into six broad racial categories: *Asian*, *Indian*, *Black*, *White*, *Middle Eastern*, and *Latino/Hispanic*. Rather

than assigning a fixed label, the model produces a probability distribution over the six categories, capturing the degree of uncertainty in racial appearance.

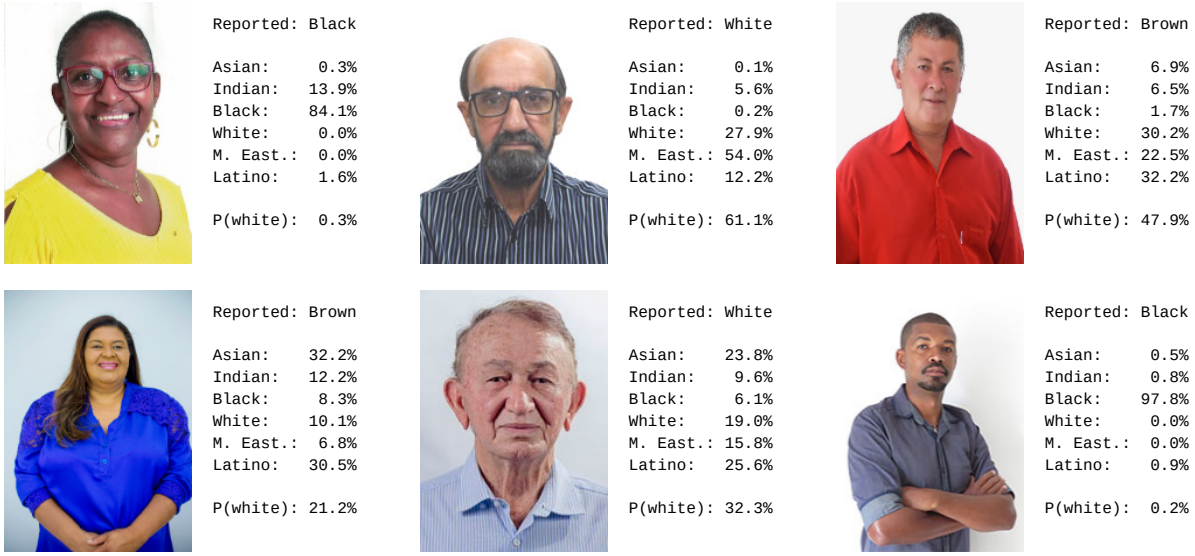

Figure 1: **Random candidate examples.** Six randomly selected candidates are shown with their self-reported race and the full probability vector produced by the DeepFace classifier. The examples illustrate the probabilistic nature of the phenotypic signal: even when the model assigns high probability to a single category, nontrivial mass often remains on adjacent categories, while ambiguous faces display relatively diffuse probability vectors.

Figure 1 provides a concrete sense of how the algorithm maps facial features into probabilistic race classifications in our setting for a random sample of candidates. For each candidate, we present the self-reported race, the full probability vector produced by the DeepFace classifier, and the predicted probability of being white from a logit model $\widehat{p}^{\text{white}}$, to be introduced in details later. The figure shows that the output is a continuous measure rather than a hard label and motivates our use of the full probability vector as a multidimensional appearance metric in the analysis.

The DeepFace pipeline follows four steps: detection, alignment, representation, and inference. Detection and alignment prepare the image by locating and normalizing facial landmarks (eyes, nose), producing a standardized numerical vector. We use the *RetinaFace* detection architecture (Deng, Guo, Zhou, Yu, Kotsia and Zafeiriou, 2019) as our default, which outperforms the alternative *OpenCV* detector (Alekhin, Smorkalov and Pisarevsky, 2026) in our validation exercise. Representation relies on VGGFace (Parkhi, Vedaldi and Zisserman, 2015), a deep con-

volutional neural network trained on 2.6 million images. Inference compares the representation to the FairFace dataset (Karkkainen and Joo, 2021), which contains 86,000 training images labeled by race/ethnicity. These models do not identify race in a biological or sociological sense; instead, they learn visual patterns—facial morphology, skin tone, texture—that co-occur with racial categories in the training data.

Performance. Serengil and Ozpinar (2024) benchmark multiple face detection and representation algorithms against human performance using the Labeled Faces in the Wild dataset (Huang, Mattar, Berg and Learned-Miller., 2008). While humans recognize faces with 97.5% accuracy in context-free images (Kumar, Berg, Belhumeur and Nayar, 2009), VGGFace achieves 96.6% accuracy in the same setting. More relevant for our purposes, Fournier-Montgieux, Borgne, Popescu and Luvison (2025) test DeepFace's demographic attribute inference for accuracy, fairness, and robustness across three balanced datasets. DeepFace achieves 79.66% average accuracy in ethnicity classification, compared with 86.07% for the alternative FairFace algorithm. DeepFace also exhibits substantially less bias across demographic groups, highlighting the accuracy-fairness trade-off noted by Fournier-Montgieux et al. (2025).

Advantages of our data. A key concern with facial classifiers is that classifications may vary across different images of the same individual due to pose, illumination, and image quality (Fournier-Montgieux et al., 2025). These factors are considerably less problematic in our setting. Candidate photographs must meet strict regulations established by the Brazilian Electoral Court: 161×225 pixel dimensions, 24-bit color depth, uniform background, and frontal pose with neutral framing.[3] These requirements substantially reduce variation from non-frontal angles, shadows, poor lighting, and background noise—sources that typically degrade algorithmic performance. The standardized, context-free environment is precisely where facial analysis systems perform best.

[3]Resolution No. 23,609 of December 19, 2019 (Art. 27, §2) specifies that photographs must have 161×225 pixel dimensions, 24-bit color depth, a uniform background, and depict the candidate in a frontal pose with appropriate attire. Ethnic or religious clothing and necessary accessories for persons with disabilities are permitted, while decorative or campaign-related elements are prohibited.

Limitations. The racial categories used by the algorithm are broad and U.S.-centric, which do not fully capture Brazil's racial context—particularly the *pardo* category, which has no direct equivalent. However, for our purposes this is not a drawback. We do not aim to definitively classify individuals' race but rather to capture their phenotypic appearance as perceived by others. Using the full probability distribution enables us to construct a multi-dimensional measure: individuals with similar probability vectors likely share similar physical traits, regardless of how they would be labeled in any particular classification system. Crucially, the algorithm functions as a *phenotypic similarity metric* rather than a racial classifier. Context-dependent racial perception—whereby a face might be categorized differently by Brazilian versus U.S. observers—would be a concern if we applied the algorithm's labels directly as a measure of race. Since our outcome is self-reported race in Brazil and the probability vector enters only as a predictor of that outcome, what matters is that individuals with similar vectors look alike, which is guaranteed by the algorithm's design independent of how the categories are named.

Validation. Since race categories in our dataset do not align with DeepFace categories, we cannot directly test accuracy. However, we exploit the fact that many candidates contested both the 2020 and 2024 elections to assess robustness—the consistency of classifications across different photographs of the same individual. Following Fournier-Montgieux et al. (2025), who use entropy-based measures in their robustness tests, we compare probability distributions using the normalized Jensen-Shannon Divergence:

$$JSD(\mathbf{z}_{i,1} \| \mathbf{z}_{i,2}) = \frac{1}{2\ln 2}\left(\sum_{r \in C} z^r_{i,1} \ln \frac{2z^r_{i,1}}{z^r_{i,1} + z^r_{i,2}} + \sum_{r \in C} z^r_{i,2} \ln \frac{2z^r_{i,2}}{z^r_{i,1} + z^r_{i,2}} \right),$$

where $\mathbf{z}_{i,p}$ is the six-dimensional probability vector with typical element $z^r_{i,p}$ for individual i from photograph p, and C is the set of race categories. The prefactor $\frac{1}{2\ln 2}$ normalizes the divergence to the unit interval. The JSD is bounded between zero (identical distributions) and one (maximally different).

We apply this measure to the set of candidates who ran in both the 2020 and 2024 municipal

elections. A total of 169,216 candidates contested both elections. Of these, *RetinaFace* successfully detected faces in both photographs for 166,604 individuals (98.46%), while *OpenCV* detected faces in both photographs for 157,744 individuals (93.22%). Every candidate whose face was detected by *OpenCV* was also detected by *RetinaFace*, confirming the latter's superior detection coverage. We compute the JSD for each individual under both algorithms and present the results in Figure 2.

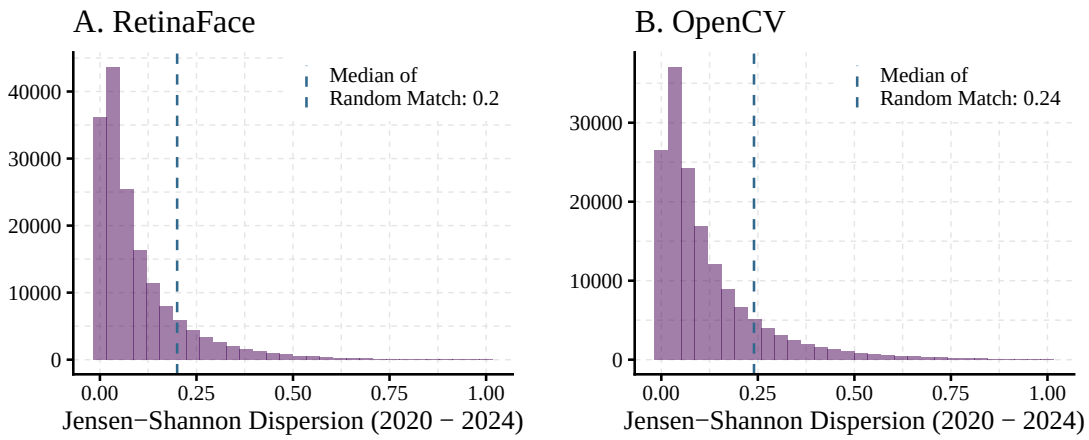

Figure 2: **Robustness of phenotypic classification.** Distribution of the Jensen-Shannon Divergence (JSD) between the probability vectors assigned to the same individual's 2020 and 2024 photographs. Panel A uses RetinaFace for face detection; Panel B uses OpenCV. The dashed vertical line indicates the median JSD obtained from 50 random permutations, in which each candidate's 2020 photograph is matched to a randomly selected 2024 photograph from a different candidate.

The median JSD is 0.056 for *RetinaFace* and 0.072 for *OpenCV*, confirming that *RetinaFace* produces more stable classifications across different photographs of the same individual. Both values are far below the theoretical maximum of one, indicating that the algorithm assigns similar probability distributions to the same person's photographs taken four years apart. To benchmark these figures, we construct a counterfactual by randomly permuting the 2024 photographs across candidates and recomputing the JSD. The median JSD under random matching is 0.20 for *RetinaFace* and 0.24 for *OpenCV*. These random-match values can be interpreted as reflecting the overall degree of phenotypic dispersion in our sample: they measure the expected divergence between probability vectors drawn from two unrelated individuals. In both samples, approximately 85% of the true matched JSD values fall below the median of the random-match distribution,

indicating that the vast majority of individuals receive substantially more similar classifications across their own photographs than would be expected from comparing two arbitrary candidates.

To further assess the reliability of the RetinaFace classifications at the category level, we decompose the total variation in each race probability into between-person and within-person components. Let $z^r_{i,t}$ denote the probability assigned to race category r for individual i in election $t \in \{2020, 2024\}$. For each category r, define the individual mean $\bar{z}^r_i = [z^r_{i,2020} + z^r_{i,2024}]/2$ and the grand mean $\bar{z}^r = (1/n) \sum_i \bar{z}^r_i$. The between-person mean square measures the dispersion of individual average classifications around the grand mean:

$$\mathrm{MSB}^r = \frac{2}{n-1} \sum_{i=1}^{n} (\bar{z}^r_i - \bar{z}^r)^2.$$

The factor of 2 arises because each individual contributes two observations, so the between-person sum of squares is $\sum_i 2(\bar{z}^r_i - \bar{z}^r)^2$; dividing by $n - 1$ gives the between-person mean square, analogous to the standard sample variance. The within-person mean square measures how much each individual's classifications vary across the two photographs:

$$\mathrm{MSW}^r = \frac{1}{n} \sum_{i=1}^{n} \sum_{t} (z^r_{i,t} - \bar{z}^r_i)^2.$$

Here, the denominator is n because each of the n individuals contributes one degree of freedom (two observations minus one mean). We define the within-person noise share as

$$W^r = \frac{\mathrm{MSW}^r}{\mathrm{MSB}^r + \mathrm{MSW}^r},$$

which is bounded in $[0, 1]$. Values close to zero indicate that nearly all variation in the algorithm's probabilities reflects stable between-person differences, with little noise introduced by differences across photographs. Figure 3 reports the results. All six categories exhibit low noise shares, confirming that the algorithm captures persistent individual-level phenotypic characteristics. This complements the JSD analysis by providing a category-level assessment: the within-person vari-

ation induced by using different photographs is small relative to the meaningful between-person variation in phenotypic appearance.

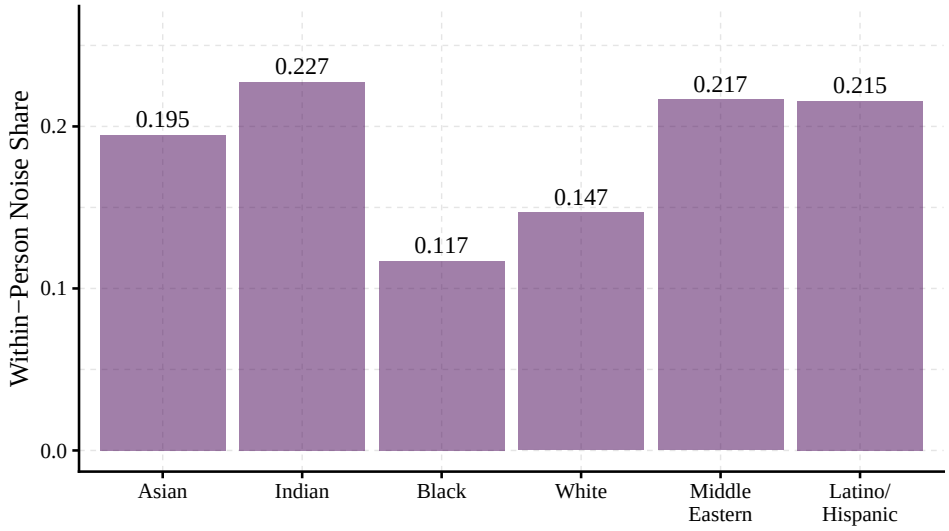

Figure 3: **Within-person noise share.** Share of total dispersion in each race classification probability attributable to within-person variation across photographs (W^r), computed from RetinaFace classifications of candidates observed in both the 2020 and 2024 elections. Lower values indicate that between-person variation dominates.

Figure 4 provides a visual complement by plotting, for each race category, the 2020 classification probability against the 2024 probability. To avoid overplotting, observations are averaged within narrow bins of the 2020 probability. If the algorithm were perfectly consistent, all points would lie on the 45-degree line. Each panel reports the slope and R^2 from a simple linear regression of the 2024 probability on the 2020 probability at the individual level. The estimated slopes range from 0.53 (*Indian*) to 0.81 (black), and R^2 values range from 0.30 to 0.59. Slopes below one indicate regression to the mean at the individual category level: extreme values in 2020 tend to be less extreme in 2024. This is expected, as variation can shift across categories for the same individual—for instance, a candidate's white probability may decrease while their *Latino/Hispanic* probability increases—without the overall probability vector changing substantially. The low JSD and noise share values reported above confirm that the joint distribution is considerably more stable than any individual marginal. Since our analysis uses the full set of six probabilities rather than

any single category, the stability of the joint vector is the relevant metric.

Because the 2020 and 2024 category means are nearly identical, the OLS fit crosses the 45-degree line close to the mean of each distribution. The discrepancy between the fitted line and the 45-degree line therefore grows toward the tails, where fewer candidates are located and photo-specific noise has more scope to push probabilities to extreme values. The algorithm is thus most reliable in the region of the probability space where most of the data is concentrated.

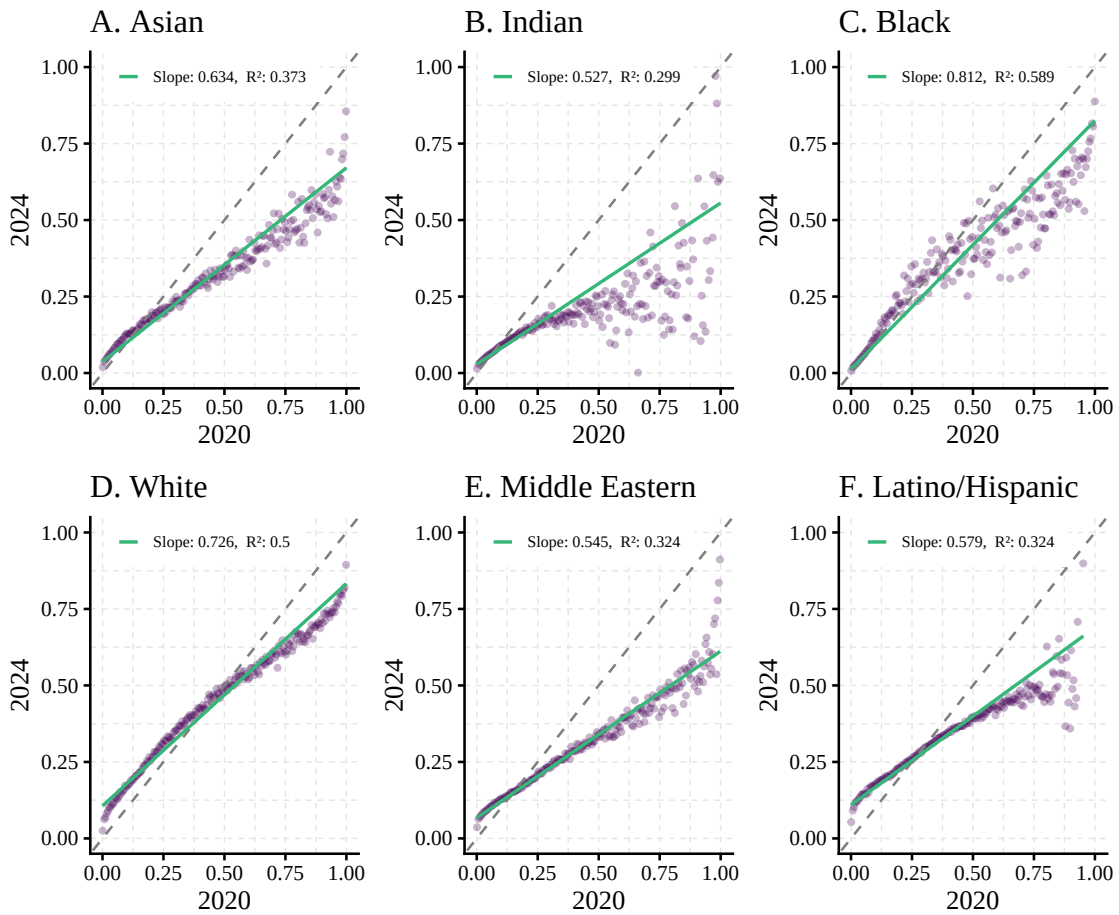

Figure 4: **Stability of race classification probabilities across elections.** Each panel plots the RetinaFace classification probability in 2020 against the probability in 2024 for one race category. Points are bin averages (bins of 0.5 percentage points on the 2020 axis). The dashed line is the 45-degree line. The legend reports the slope and R^2 from a linear regression of the 2024 probability on the 2020 probability at the individual level.

Finally, we examine whether the algorithm's classification probabilities are meaningfully related to self-reported race. Figure 5 plots the share of candidates who self-identify as white against the

algorithm's predicted probability of being classified as white, computed in bins of 5 percentage points. The relationship is strongly monotonic: candidates whom the algorithm classifies with a higher white probability are substantially more likely to self-identify as white. The relationship lies above the 45-degree line at low predicted probabilities and below it at high probabilities, consistent with the patterns documented in Section 4, where social context and socioeconomic factors shape self-identification beyond appearance alone—individuals with ambiguous appearance are more likely to self-identify as white than a purely appearance-based classification would predict, while those with clearly white appearance self-identify as such at rates below 100%, reflecting individuals who identify as *pardo* or other categories despite a phenotypically white appearance.

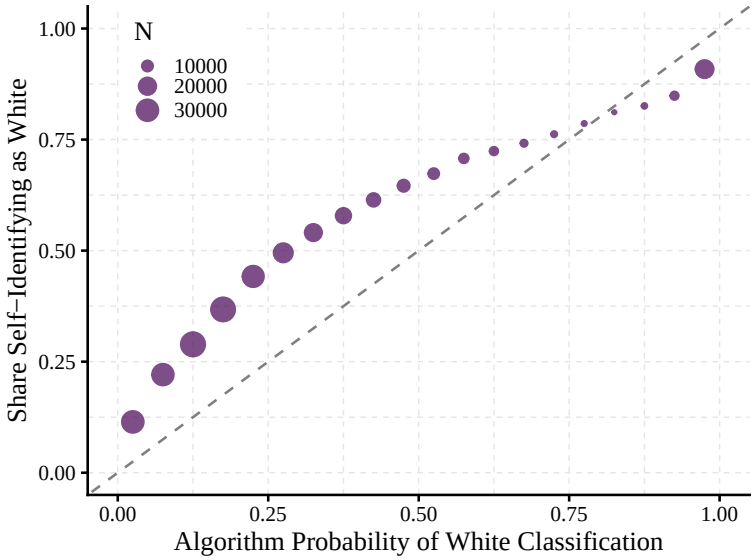

Figure 5: **Calibration of phenotypic classification.** Share of candidates self-identifying as white by the algorithm's predicted probability of white classification (RetinaFace, 5 percentage point bins). Point size is proportional to the number of candidates in each bin. The dashed line represents perfect calibration.

Taken together, these results support the use of the DeepFace pipeline as a reliable tool for measuring phenotypic appearance in our setting. The standardized photograph requirements imposed by the Brazilian Electoral Court substantially reduce the sources of variation that typically degrade algorithmic performance, and the resulting classifications exhibit a high degree of consistency across election cycles. *RetinaFace* outperforms *OpenCV* on both detection coverage and robust-

ness, which motivates our choice of *RetinaFace* as the default face-detection algorithm throughout the analysis, with *OpenCV* results reported for robustness.

3.4 Phenotypic Ambiguity

The probability vector $\mathbf{z}_i$ is six-dimensional, but for the study of racial self-identification the relevant dimension is the *white/nonwhite* axis: how likely is individual i to be perceived as white in the Brazilian social context? Projecting $\mathbf{z}_i$ onto this single axis requires care. The raw probability of white classification, z_i^{white}, does not account for the fact that other categories—particularly *Middle Eastern* and *Asian*—are typically perceived as non-black and often read as white in Brazil. An individual with $z_i^{\text{white}} = 0.50$ and $z_i^{\text{Middle Eastern}} = 0.50$ would appear maximally ambiguous under a measure based on z_i^{white} alone, despite being unambiguously perceived as white (or at least non-black) in the Brazilian context.

To address this, we construct a scalar index of perceived whiteness by projecting the full appearance vector onto the Brazilian white/nonwhite classification axis. Specifically, we regress self-reported white status on the full probability vector using a logistic model and use the fitted values as the projected index:

$$\widehat{p}_i^{\text{white}} \;=\; \sigma\left(\widehat{\boldsymbol{\beta}}^{\top}\mathbf{z}_i^{-}\right),$$

where $\sigma(\cdot)$ is the logistic function, $\mathbf{z}_i^{-}$ is the five-element subvector of $\mathbf{z}_i$ excluding *Latino/Hispanic* (omitted as reference category to avoid perfect multicollinearity), and $\widehat{\boldsymbol{\beta}}$ is estimated by logistic regression of y_i on $\mathbf{z}_i^{-}$ using the full panel of 2020 and 2024 candidates. Crucially, $\widehat{p}_i^{\text{white}}$ is used only to construct the ambiguity measure; the Shapley-Owen decomposition and all regression analyses use the raw appearance probabilities $\mathbf{z}_i$ as regressors, so the in-sample logit fit does not enter the estimation of treatment effects or variance shares. The fitted values $\widehat{p}_i^{\text{white}}$ serve as a *sufficient statistic* for the appearance vector along the white/nonwhite classification axis: conditional on $\widehat{p}_i^{\text{white}}$, the remaining components of $\mathbf{z}_i$ carry no additional information for predicting self-reported white status. This projection weights each DeepFace category by its empirical relevance for racial

self-identification in Brazil, rather than by the algorithm's U.S.-centric labeling convention.

Figure 6 plots the distribution of $\widehat{p}_i^{\text{white}}$ separately for candidates who self-declared as white, brown (*pardo*), and black. The measure passes a basic face-validity test: white self-identifiers have substantially higher $\widehat{p}_i^{\text{white}}$ than black self-identifiers, with brown candidates occupying an intermediate position. Notably, the average self-identified white candidate has $\widehat{p}_i^{\text{white}} \approx 0.70$ rather than values close to one, reflecting the fact that many self-identified white Brazilians do not present as unambiguously white to an algorithm trained on an internationally diverse dataset—a pattern consistent with the social whitening documented in the analysis.

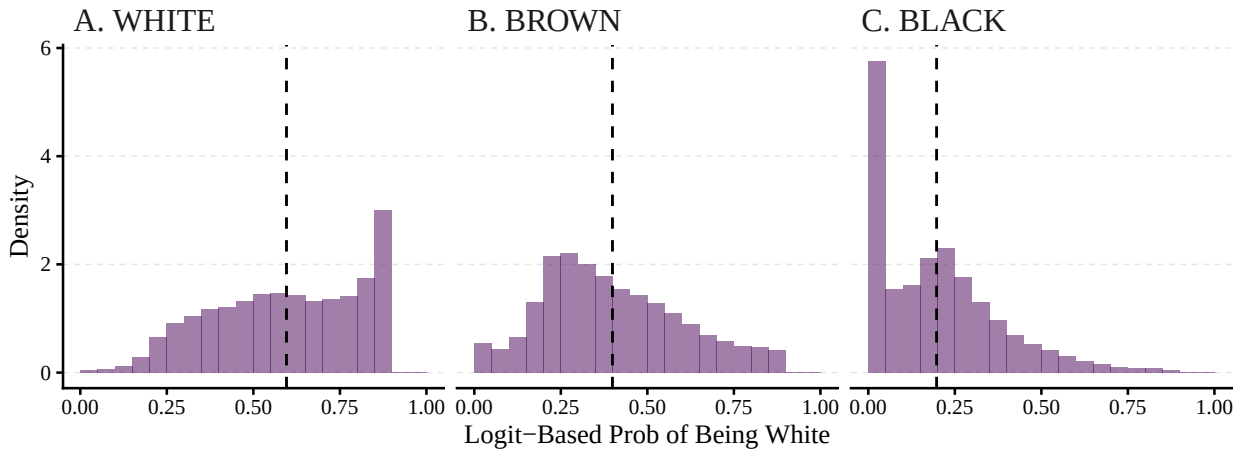

Figure 6: **Distribution of perceived whiteness by self-reported race.** Distribution of $\widehat{p}_i^{\text{white}}$—the logit-based probability of white self-identification implied by the DeepFace appearance vector—for candidates self-declaring as *branca* (white), *parda* (brown/mixed-race), and *preta* (black) in the 2020 elections. Dashed lines indicate group means. The measure correctly orders the three groups along the white/nonwhite axis and reveals substantial within-group dispersion, particularly among white and brown self-identifiers.

In the heterogeneity analysis, we stratify candidates by $A_i = 1 - 2|\widehat{p}_i^{\text{white}} - 0.5|$ and examine whether the effects of socioeconomic characteristics and historical slavery on white self-identification are stronger for high-ambiguity individuals. We verify robustness of this analysis to two alternative moderators: the raw index $\widehat{p}_i^{\text{white}}$ (without the ambiguity transformation), and the white-centered ambiguity $1 - 2|z_i^{\text{white}} - 0.5|$ based on the raw DeepFace white probability rather than the logit-projected index.

3.5 Socioeconomic Characteristics

The set of individual socioeconomic variables is obtained from the candidacy information from the Brazilian Electoral Court (TSE). Candidates report their race, gender, marital status, occupation, educational attainment, and age. The reported race serves as our outcome variable. Although candidates may identify as white (*branca*), black (*preta*), *yellow* (*amarela*), brown (*parda*), or *indigenous* (*indígena*), we construct a binary variable indicating whether the candidate reported being white (1) or nonwhite (0). All other variables are included in the vector $\mathbf{x}_i$ of socioeconomic characteristics. Candidates are additionally required to declare the value of their assets, which we use in robustness checks. These declarations are self-reported and unverified; many candidates report zero wealth, likely reflecting underreporting rather than true asset absence (Souto-Maior and Borba, 2019). Appendix E presents the analysis on the subsample of candidates with positive asset declarations.

In most estimations, we select *Latino/Hispanic* as the reference category for race classification. *Female* is the reference for gender, and *Less than primary education* is the reference for education. For marital status, we group *Separated*, *Divorced*, and *Widowed* individuals as *Previously Married*, which serves as the reference category. Occupation is reported using the TSE's detailed classification system, which we aggregate into 10 broad sectoral groups (e.g., professional, educator, public servant, agricultural; see Appendix A for the full list). Occupation group indicators are included in all regressions with *other* (retirees, students, religious figures, artists, media professionals, and undisclosed) as the reference category.

3.6 Historical and Other Controls

To capture the historical legacy of slavery, we use data from the 1872 Brazilian Census, which provides the Share of Enslaved Population (1872), the ratio of enslaved individuals to the total population in each minimum comparable area of 1872. This variable is matched to modern municipalities using historical boundary correspondence. Geographic controls include the median temperature, median elevation, median rainfall, median terrain ruggedness index, log distance

to rivers, log distance to the coast, and log area size of each municipality. Municipality-level controls, derived from the 2022 Population Census (IBGE), include the share of white individuals, average years of schooling, and average wage by race. The land Gini coefficient is computed from the 2017 Agricultural Census, measuring land inequality within each municipality. Geographic controls are constructed from shapefiles and rasters from Geobr (Pereira, Gonçalves et al., 2019) and the National Institute of Spatial Research. Crop suitability is measured using rasters from FAO GAEZ (FAO/IIASA, 2012), while data on number of indigenous tribes is from the seminal work of Nimuendajú (2017) georeferenced by Barsanetti and Ferreira (2024).

4 Determinants of Racial Identity

Table 1 presents regression results examining how phenotypic appearance, as inferred from facial features, relates to self-identified racial classification. The dependent variable is a binary indicator equal to one if the individual reported being white. The regressors are the algorithm-assigned probabilities that the individual belongs to each racial category, with *Latino/Hispanic* as the reference. Columns (1)–(3) use the 2020 cross-section; columns (4)–(6) use the panel sample with appearance probabilities averaged across the 2020 and 2024 photographs. Within each sample, we progressively add socioeconomic controls and municipality fixed effects.

The probability of being algorithmically classified as white is a strong predictor of self-identifying as white. In the baseline specification without controls (column 1), the coefficient of 0.702 implies that a 10-percentage-point increase in the predicted probability of looking white is associated with a 7.0-percentage-point increase in the likelihood of self-identifying as white. This effect attenuates when socioeconomic controls and municipality fixed effects are added (column 3: 0.442), but remains highly significant, indicating that appearance remains a strong predictor of racial self-identification even after controlling for individual characteristics and local context.

The coefficients on the other appearance categories reveal systematic patterns. The probability of self-identifying as white increases with a *Middle Eastern* appearance (coefficient of 0.305–0.537

Table 1: **Determinants of Racial Self-Identification.**

	2020 Cross-Section			Panel (avg. photos)		
	(1)	(2)	(3)	(4)	(5)	(6)
Dependent Variable: Self-Reported White						
z^{White}	0.702***	0.669***	0.442***	0.818***	0.782***	0.500***
	(0.006)	(0.006)	(0.005)	(0.009)	(0.010)	(0.009)
z^{Asian}	0.107***	0.091***	0.075***	0.217***	0.189***	0.134***
	(0.007)	(0.007)	(0.006)	(0.012)	(0.012)	(0.011)
z^{Black}	-0.255***	-0.257***	-0.345***	-0.205***	-0.214***	-0.356***
	(0.007)	(0.007)	(0.010)	(0.010)	(0.010)	(0.012)
z^{Indian}	-0.619***	-0.602***	-0.538***	-0.854***	-0.844***	-0.806***
	(0.014)	(0.014)	(0.012)	(0.026)	(0.026)	(0.023)
$z^{\text{Middle Eastern}}$	0.424***	0.408***	0.305***	0.537***	0.515***	0.359***
	(0.007)	(0.007)	(0.006)	(0.013)	(0.013)	(0.011)
Socioeconomic Controls		✓	✓		✓	✓
Municipality FE			✓			✓
Observations	520,733	520,733	520,733	169,175	169,175	169,175
N Clusters	5,568	5,568	5,568	5,567	5,567	5,567
R^2	0.231	0.245	0.402	0.259	0.271	0.440
Adjusted R^2	0.231	0.245	0.396	0.259	0.271	0.421

Notes: The dependent variable is self-reported white (=1). All appearance probabilities are from the DeepFace algorithm; Latino/Hispanic is the reference category. Socioeconomic controls include gender, marital status, educational attainment, log age, and occupation group indicators (see Appendix A). Standard errors clustered at the municipality level in parentheses. Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

depending on specification and sample), suggesting that social perceptions in Brazil tend to align Middle Eastern features with whiteness. In contrast, black and *Indian* appearances are strongly negatively associated with white self-identification, with *Indian* showing the largest negative effect (−0.538 to −0.854). The large negative coefficient for *Indian*—which in the DeepFace taxonomy refers to South Asian appearance—likely reflects that the visual features the algorithm associates with this category, such as darker skin tone, overlap with phenotypic traits of Brazilians who identify as brown or black. The coefficient on *Asian* is positive but modest (0.075–0.107), consistent with the fact that individuals of Asian descent in Brazil may self-identify as white, *yellow*, or *pardo*.

The R^2 in column (1) is 0.231, indicating that, in a linear model, appearance alone explains about 23 percent of the variation in self-reported race. Adding socioeconomic controls in column (2) increases the R^2 only marginally to 0.245, while the inclusion of municipality fixed effects in column (3) raises it substantially to 0.402. This pattern suggests that local social context—captured by municipality fixed effects—plays a role comparable to physical appearance in shaping racial identity, whereas individual socioeconomic characteristics contribute relatively little. The panel sample (columns 4–6) yields qualitatively similar results, with slightly larger coefficients on appearance probabilities, consistent with the reduction in measurement error from averaging photographs across elections. These results are robust to estimating logit models, which correctly account for the binary nature of the dependent variable; the pattern of coefficients and the relative contributions of each component to model fit are preserved (see Appendix B).

Decomposing explained variation. The sequential comparison of R^2 across columns of Table 1 is informative but order-dependent: the marginal contribution of each group of variables depends on which controls are already included. To obtain an order-invariant decomposition, we apply the Shapley-Owen method (Shorrocks, 2013; Israeli, 2007), which assigns each group of regressors its average marginal contribution to adjusted R^2 across all possible orderings of entry. With three groups—appearance probabilities ($\mathbf{z}_i$), socioeconomic characteristics ($\mathbf{x}_i$), and municipality fixed effects (w_i)—there are $2^3 - 1 = 7$ non-empty subsets and $3! = 6$ orderings. The Shapley value for

group g is

$$\phi_g = \frac{1}{3!} \sum_{\pi \in \Pi} \left[\bar{R}^2(S_g^\pi \cup \{g\}) - \bar{R}^2(S_g^\pi) \right], \tag{2}$$

where Π is the set of all orderings of the three groups and S_g^π denotes the set of groups that precede g in ordering π. The three Shapley values sum exactly to the full-model adjusted R^2, providing an additive decomposition into the share of explained variation attributable to each component.

Panel A of Figure 7 presents the results. Each bar reports the Shapley value in adjusted R^2 units—the share of total variation in self-reported race attributable to each component. Municipality fixed effects account for the largest share (0.209), followed by appearance probabilities (0.166), with socioeconomic characteristics contributing an order of magnitude less (0.021).[4] The dominance of municipality fixed effects confirms that local social context plays the largest role in shaping racial self-identification, while the substantial contribution of appearance validates the algorithm's ability to capture meaningful phenotypic information. The modest value attributed to socioeconomic variables suggests that, conditional on how individuals look and where they live, factors such as education, age, and wealth play a comparatively minor role in racial self-identification.

Heterogeneity by racial ambiguity. The aggregate decomposition masks important variation across individuals with different degrees of phenotypic ambiguity. To investigate how the relative importance of each component changes along this dimension, we repeat the Shapley-Owen decomposition within subgroups defined by the logit-based ambiguity measure A_i introduced in Section 3.4. We partition the sample into quintiles of A_i and compute the Shapley-Owen decomposition within each bin.[5]

Panel B of Figure 7 presents the results as stacked bars, where each bar's total height equals the

[4]The three groups have asymmetric numbers of parameters: appearance uses 5 probabilities, socioeconomic characteristics use approximately 15 variables, and municipality fixed effects use approximately 5,500 parameters. A larger number of parameters mechanically increases the potential fit, which could inflate the Shapley contribution of the fixed-effect group. We therefore treat the ordering—municipality fixed effects > appearance ≫ socioeconomic characteristics—as the robust finding, rather than exact magnitudes. This pattern is confirmed by the logit pseudo-R^2 decomposition (Appendix B) and the two-group decomposition excluding appearance.

[5]Each quintile contains approximately 104,000 candidates (total $N = 520{,}733$). The approximate quintile midpoints of A_i (used as evaluation points in the linear interaction figures) are $A \approx 0.17, 0.42, 0.60, 0.76, 0.92$ for Q1–Q5, respectively. Appendix B presents robustness using the white-centered measure $1 - 2|z_i^{\text{white}} - 0.5|$, the panel sample, and the raw $\widehat{p}_i^{\text{white}}$ probability, all of which produce nearly identical patterns.

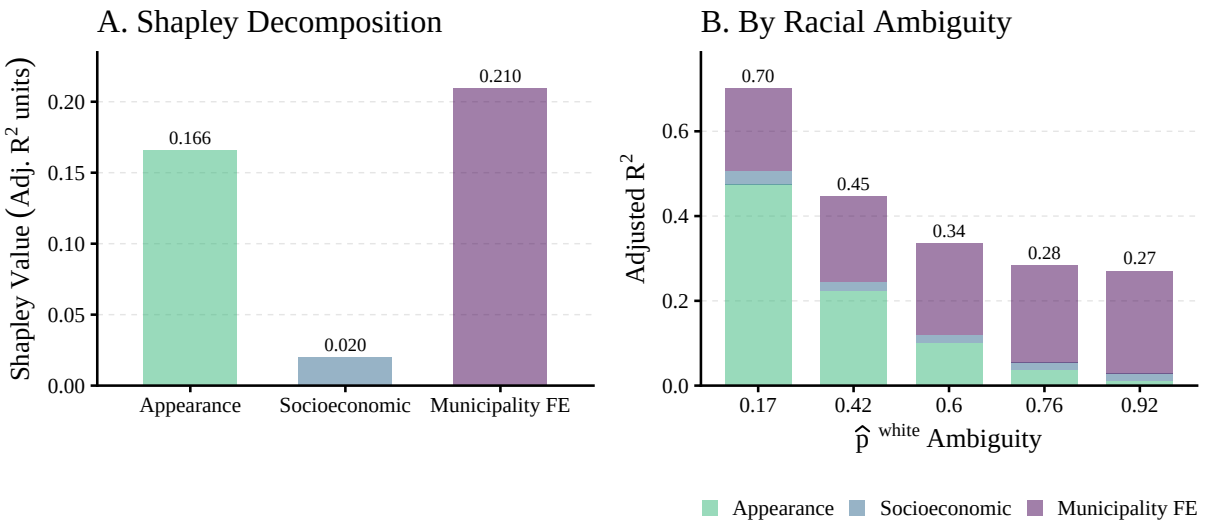

Figure 7: **Shapley-Owen decomposition of adjusted R^2.** Panel A shows Shapley values in adjusted R^2 units for three groups of regressors: appearance probabilities, socioeconomic characteristics, and municipality fixed effects. The decomposition is computed by averaging each group's marginal contribution to adjusted R^2 across all possible orderings of entry. Panel B repeats the decomposition within quintiles of logit-based racial ambiguity $A_i = 1 - 2|\widehat{p}_i^{\text{white}} - 0.5|$ (see Section 3.4), where higher values indicate greater phenotypic ambiguity about whiteness. Each stacked bar's total height equals the full-model adjusted R^2 for that quintile; colored segments show each component's contribution.

full-model adjusted R^2 for that ambiguity quintile, and the colored segments show each component's Shapley contribution. The pattern reveals that appearance explains less and less variation as racial ambiguity increases: its Shapley value—the order-invariant average contribution to adjusted R^2—falls from 0.475 in the lowest ambiguity quintile to just 0.011 in the highest. This decline is expected—when an individual's appearance does not clearly signal whiteness, appearance probabilities lose predictive power for self-identification.

This pattern reveals that the relative importance of social constructs rises sharply with ambiguity: the share of explained variation attributable to municipality fixed effects increases from 28% in the least ambiguous quintile to 89% in the most ambiguous. However, this relative increase is driven almost entirely by the collapse in appearance's contribution, not by large gains in the absolute explanatory power of social factors. Municipality fixed effects increase only modestly in absolute terms (from 0.195 to 0.243), while socioeconomic characteristics change little (from 0.032 to

0.018). The total explained variation itself falls substantially (from adjusted $R^2 = 0.70$ to adjusted $R^2 = 0.27$) as we move toward more ambiguous individuals.

These findings reinforce the view that social construction of racial identity operates primarily at the margin of phenotypic ambiguity. For individuals whose appearance unambiguously signals a racial category, self-identification is largely determined by how they look. But for those with racially ambiguous appearances—where the algorithm assigns similar probabilities to being white or not—the community in which they live becomes the dominant factor shaping how they report their race. Socioeconomic characteristics, by contrast, contribute modestly regardless of where individuals fall on the ambiguity spectrum. The municipality fixed effects capture a mixture of compositional and normative channels: local racial composition (the share of white residents) and local norms are both absorbed into the fixed effects. The mediation analysis in Section 7 shows that approximately 54% of the slavery–self-identification association operates through contemporary racial composition, suggesting that a substantial fraction of the fixed effects's explanatory power reflects demographic facts rather than purely normative environments.

These results are robust to alternative specifications. In Appendix B, we show that the pattern replicates using the white-centered ambiguity measure $1 - 2|z_i^{\text{white}} - 0.5|$, which produces a nearly identical gradient. We also verify robustness in the panel sample using averaged photographs, where appearance contributions decline from 0.479 to 0.040 and municipality fixed effects rise from 0.209 to 0.239 across $\widehat{p}_i^{\text{white}}$ ambiguity quintiles. A two-group decomposition excluding appearance shows that municipality fixed effects account for nearly all explained variation (0.263 of 0.291) when phenotypic information is omitted, confirming that local social context is the dominant non-appearance determinant of racial identity. Plotting effects directly against the raw $\widehat{p}_i^{\text{white}}$ addresses the floor/ceiling concern for the lowest-ambiguity quintile by separating clearly-white from clearly-nonwhite individuals, confirming that the small effects at low ambiguity reflect binding constraints at both extremes.

Measurement error. A natural concern is that the decline in appearance's explanatory power reflects increased measurement error for individuals with ambiguous appearances, rather than a genuine reduction in the role of phenotype. Under classical measurement error, if we observe $v = v^* + u$ instead of the true appearance v^*, where u is noise with variance σ_u^2, the OLS coefficient is attenuated by the reliability ratio $\lambda = \mathrm{Var}(v^*)/[\mathrm{Var}(v^*) + \sigma_u^2]$. The contribution to R^2 scales proportionally with λ: if measurement error variance doubles, the reliability ratio falls, but by a factor that depends on the initial signal-to-noise ratio—i.e., the ratio of true variation in appearance, $\mathrm{Var}(v^*)$, to measurement noise, σ_u^2. With high reliability (low initial noise), a doubling of σ_u^2 might reduce λ from 0.95 to 0.90—a 5% decline, not the 96% decline (or equivalently, a 25-fold reduction) we observe in appearance's Shapley contribution.[6]

To assess whether differential measurement error could explain our findings, we stratify the Jensen-Shannon Divergence (JSD) between 2020 and 2024 photographs by ambiguity quintile. If measurement error were substantially higher for ambiguous individuals, we would expect their JSD values to be correspondingly elevated. We find that median JSD approximately doubles from the lowest to the highest ambiguity quintile. This indicates that the algorithm's classifications are somewhat less stable for individuals with ambiguous appearances, as one might expect when faces are harder to classify.

However, this twofold increase in measurement error is unlikely to account for the 25-fold reduction in appearance's contribution to explained variation. We therefore conclude that the decline in appearance's explanatory power primarily reflects a genuine reduction in the role of phenotype for self-identification among individuals whose appearance does not clearly signal a racial category. The overall adjusted R^2 also declines as ambiguity increases—from 0.70 in the lowest quintile to 0.27 in the highest—indicating that unobserved factors account for a larger share of the variation in racial self-identification when appearance is ambiguous. Note, however, that part

[6]Formally, if $v = v^* + u$ with $u \perp v^*$, then $R^2(y \sim v) = \lambda \cdot R^2(y \sim v^*)$ where $\lambda = \sigma_{v^*}^2/(\sigma_{v^*}^2 + \sigma_u^2)$ is the reliability ratio. When noise doubles ($\sigma_u^2 \to 2\sigma_u^2$), the new reliability is $\lambda' = \sigma_{v^*}^2/(\sigma_{v^*}^2 + 2\sigma_u^2)$, so the R^2 contribution falls by a factor $\lambda/\lambda' = (1 + 2r)/(1 + r)$, where $r = \sigma_u^2/\sigma_{v^*}^2$. Since $(1 + 2r)/(1 + r) = 1 + r/(1 + r) < 2$ for all finite r, a twofold increase in noise can reduce the R^2 contribution by strictly less than a factor of 2, regardless of the initial signal-to-noise ratio.

of this decline is mechanical: as ambiguity increases, the sample's white self-identification rate approaches 50%, increasing the variance in the outcome and mechanically lowering R^2 regardless of any increase in truly unobserved heterogeneity. Both forces are likely at play. Nevertheless, within the explained variation, municipality fixed effects gain relative importance: their absolute contribution rises modestly (from 0.191 to 0.254) even as appearance's contribution collapses. This pattern—appearance losing explanatory power while local context retains or slightly increases its role—is consistent with our interpretation that social environment matters most precisely when phenotype provides weak signals.

Unobserved determinants. The decline in overall adjusted R^2 for individuals with ambiguous appearances raises the question of what unobserved factors shape their racial self-identification. Several factors merit consideration. First, family racial identity and ancestry narratives likely play an important role: how parents and grandparents identify, stories about European or African ancestors, and the racial composition of the extended family may all influence how individuals with ambiguous phenotypes come to see themselves. Second, personal experiences with racial classification—being treated as white or nonwhite in schools, workplaces, or social interactions—may accumulate over time to shape identity in ways our cross-sectional data cannot capture (see Portella et al., 2022, for evidence on peer-network effects on racial identification). Third, cultural affiliations, such as participation in Afro-Brazilian religious traditions or cultural movements, may reinforce particular racial identities independent of appearance.

Fourth, finer-grained spatial variation below the municipality level—neighborhood racial composition, workplace environment, or peer networks—may matter in ways that municipality fixed effects cannot fully absorb. Finally, life events such as interracial marriage or geographic mobility may prompt individuals to reconsider their racial identity. These factors share a common feature: they operate through social interaction and lived experience rather than through phenotype or socioeconomic status, and their influence is likely strongest precisely when appearance provides ambiguous signals about racial category membership.

Despite this remaining unexplained variation, these findings establish that municipality-level factors are the dominant non-appearance determinant of racial self-identification. For individuals with racially ambiguous appearances, the community in which they live—not their education, wealth, or other individual characteristics—shapes whether they identify as white. Before investigating the historical roots of this spatial variation, we first examine the direct effects of socioeconomic characteristics on racial self-identification and whether "social whitening" operates more strongly for individuals with ambiguous phenotypes.

5 Social Whitening

We next examine the association between socioeconomic characteristics and racial self-identification. It is well documented that individuals who identify as white tend to achieve higher levels of education and income (Monk Jr., 2016; Laudares and Valencia Caicedo, 2023), largely because racial identification correlates with historical and structural differences in access to schooling and labor market opportunities. We therefore ask whether individuals with more favorable socioeconomic characteristics are also more likely to self-identify as white, conditional on their phenotypic appearance and local social context.

We interpret the associations documented below with caution: the cross-sectional design cannot rule out reverse causality, whereby individuals who identify as white face less discrimination and consequently achieve higher educational attainment or accumulate more wealth (Schwartzman, 2007; Francis and Tannuri-Pianto, 2013). A related concern is omitted background characteristics: candidates who have higher socioeconomic status for unobserved reasons—family networks, immigration generation, or neighborhood of origin—may also carry ancestry information not fully captured by the appearance vector, and those same family origins may predispose white identification. In this case, the socioeconomic-status–identity association reflects selection on unobserved background rather than a behavioral response to achieved status. Our evidence is therefore consistent with "social whitening" but does not establish a causal direction.

Table 2 presents OLS regressions with a progressive set of controls to isolate the direct effect of socioeconomic characteristics. Columns (1) and (4) include only socioeconomic variables and occupation group indicators; columns (2) and (5) add appearance controls; columns (3) and (6) further add municipality fixed effects. The R^2 progression mirrors the pattern from Table 1: socioeconomic characteristics and occupation alone explain less than 4% of the variation in racial self-identification, but this rises to 25% with appearance controls and 40% with municipality fixed effects, confirming that socioeconomic factors play a minor role relative to phenotype and local context.

The coefficient attenuation pattern reveals how socioeconomic status correlates with appearance. Consider education: without controls, candidates with tertiary education are 15.8 percentage points more likely to identify as white than those with less than primary education (column 1). Adding appearance controls reduces this to 8.2 percentage points (column 2), and municipality fixed effects leave it at 8.1 percentage points (column 3). The approximately 50% attenuation when adding appearance indicates that more educated individuals tend to have lighter phenotypes—a reflection of Brazil's historical stratification where European descendants had greater access to schooling (Rocha, Ferraz and Soares, 2017). However, the substantial residual effect of 8.1 percentage points that survives both appearance and municipality controls is consistent with "social whitening": conditional on looking the same and living in the same municipality, more educated individuals are more likely to identify as white. Similar patterns hold for age: older candidates are also more likely to identify as white.

The male coefficient exhibits a different pattern: it starts at −3.0 percentage points without controls (column 1), indicating men are less likely to identify as white, and attenuates to −1.5 percentage points with full controls (column 3). The attenuation indicates that men tend to have darker phenotypes on average, while the small but persistent residual effect suggests a minor gender difference in racial self-identification that is not fully explained by appearance or location. Here, it is important to recall that the candidate sample is not representative of the general population—it is predominantly male—such that results should be interpreted as conditional on being a political

candidate.

Marital status shows a notable pattern: both married and single individuals are less likely to self-identify as white compared to formerly married individuals, with single candidates showing the strongest effect (-10.7 percentage points without controls, -1.7 with full controls). This pattern persists across specifications, though the magnitude weakens substantially after accounting for appearance and municipality. In fact, in the panel sample with full controls (column 6), there is no significant difference between being formerly married or married.

To assess whether social whitening operates more strongly for individuals whose appearance is harder to classify, we examine how the effect of socioeconomic characteristics varies with the degree of racial ambiguity. Using the logit-based ambiguity measure A_i defined in Section 3.4, we estimate interaction models that allow the marginal effect of each socioeconomic variable to vary with ambiguity, controlling for appearance and municipality fixed effects.

Our first specification is a linear interaction model:

$$y_i = \sum_k \beta_k x_{ik} + \sum_k \gamma_k (A_i \times x_{ik}) + \boldsymbol{\pi}' \mathbf{z}_i + w_i + \varepsilon_i \tag{3}$$

where y_i is an indicator for self-identifying as white, x_{ik} are socioeconomic characteristics (education, age, gender, marital status, and occupation group indicators), $\mathbf{z}_i$ is the vector of race probabilities from the algorithm, and w_i are municipality fixed effects. In this specification, the marginal effect of characteristic k at ambiguity level A is $\Delta y / \Delta x_k = \beta_k + \gamma_k A$, which varies linearly with ambiguity.[7]

To allow for non-linear heterogeneity, we also estimate a cubic spline interaction model:

$$y_i = \sum_k \beta_k x_{ik} + \sum_k \sum_{j=1}^{3} \delta_{kj} \left(s_j(A_i) \times x_{ik} \right) + \boldsymbol{\pi}' \mathbf{z}_i + w_i + \varepsilon_i \tag{4}$$

[7]The ambiguity measure $A_i = 1 - 2|\widehat{p}_i^{\text{white}} - 0.5|$ is a nonlinear transformation of the logit-fitted probability $\widehat{p}_i^{\text{white}}$, which is itself a deterministic function of $\mathbf{z}_i$ and therefore fully absorbed once $\mathbf{z}_i$ enters the model. This dual role introduces some collinearity between $A_i \times x_{ik}$ and the linear controls in $\mathbf{z}_i$, but because A_i is a non-monotone (V-shaped) function of $\widehat{p}_i^{\text{white}}$ while the first-order controls enter linearly, the two terms are not collinear. The interaction coefficients γ_k therefore remain identified, capturing variation in the socioeconomic-status–identity association that is genuinely associated with phenotypic ambiguity rather than the level of $\widehat{p}_i^{\text{white}}$.

Table 2: **Socioeconomic Determinants of Racial Self-Identification.**

	2020 Cross-Section			Panel (avg. photos)		
	(1)	(2)	(3)	(4)	(5)	(6)
Dependent Variable: Self-Reported White						
Male	-0.030***	-0.021***	-0.015***	-0.014***	-0.011***	-0.006**
	(0.002)	(0.001)	(0.001)	(0.003)	(0.003)	(0.002)
Married	-0.040***	-0.021***	-0.008***	-0.037***	-0.019***	-0.004
	(0.003)	(0.002)	(0.002)	(0.004)	(0.004)	(0.003)
Single	-0.107***	-0.062***	-0.017***	-0.101***	-0.054***	-0.013***
	(0.004)	(0.003)	(0.002)	(0.005)	(0.004)	(0.004)
Primary Education	0.037***	0.025***	0.015***	0.032***	0.021***	0.012***
	(0.003)	(0.003)	(0.002)	(0.005)	(0.004)	(0.004)
Secondary Education	0.062***	0.036***	0.039***	0.057***	0.032***	0.034***
	(0.004)	(0.003)	(0.002)	(0.005)	(0.004)	(0.003)
Tertiary Education	0.158***	0.082***	0.081***	0.155***	0.078***	0.073***
	(0.005)	(0.004)	(0.002)	(0.006)	(0.005)	(0.004)
Age (log)	0.115***	0.088***	0.057***	0.118***	0.090***	0.050***
	(0.004)	(0.004)	(0.003)	(0.007)	(0.006)	(0.005)
Occupation Dummies	✓	✓	✓	✓	✓	✓
Appearance Controls		✓	✓		✓	✓
Municipality FE			✓			✓
Observations	520,720	520,720	520,720	169,174	169,174	169,174
N Clusters	5,568	5,568	5,568	5,567	5,567	5,567
R^2	0.037	0.245	0.402	0.033	0.271	0.440
Adjusted R^2	0.037	0.245	0.396	0.033	0.271	0.421

Notes: The dependent variable is self-reported white (=1). Reference categories: female (for Male), separated/divorced/widowed (for marital status), less than primary education (for education), other (for occupation). Appearance controls include five race probability categories from the DeepFace algorithm (with Latino/Hispanic as the excluded reference category). All specifications include occupation group indicators (coefficients not shown; see Appendix A for details). Standard errors clustered at the municipality level in parentheses. Columns (1)–(3) use the 2020 cross-section; columns (4)–(6) use the panel sample with appearance probabilities averaged across 2020 and 2024 photographs. Within each sample, the progression adds appearance controls (column 2/5) and municipality fixed effects (column 3/6) to isolate the direct effect of socioeconomic characteristics on racial self-identification. Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

where $s_1(\cdot), s_2(\cdot), s_3(\cdot)$ are natural cubic spline basis functions with two interior knots placed at the 33rd and 67th percentiles of the ambiguity distribution. The marginal effect of characteristic k at ambiguity A is now $\Delta y/\Delta x_k = \beta_k + \sum_{j=1}^{3} \delta_{kj} s_j(A)$, which can vary flexibly with ambiguity. Standard errors for these marginal effects are computed using the delta method.

Figure 8 presents the results. The points (with error bars) show marginal effects from the linear interaction model evaluated at quintile midpoints, while the curves (with shaded confidence bands) show the cubic spline estimates with 95% confidence intervals. Education effects exhibit a clear positive gradient with ambiguity: in the linear interaction model, tertiary education increases the probability of identifying as white by approximately 8 percentage points for individuals with unambiguous appearance (ambiguity ≈ 0.17), rising from 3.6 to 11.3 percentage points for those with highly ambiguous appearance (ambiguity ≈ 0.92). The spline model reveals that this relationship is non-linear—effects increase steeply until ambiguity reaches approximately 0.5, then level off. This saturation pattern suggests that social whitening has diminishing returns at extreme ambiguity, possibly because other unobserved factors (such as family narratives or cultural ties) become more important when appearance provides little signal. This pattern is consistent with Francis and Tannuri-Pianto (2013), who find that socioeconomic status has the greatest influence on racial self-classification among individuals near the boundaries between racial categories on the skin tone continuum. Similar patterns emerge for the other education levels. Age also exhibits a similar pattern, where effects increase from approximately zero to 9.6 percentage points across the ambiguity distribution but show no signs of saturation.

Our finding that education predicts white self-identification should be situated within a broader literature that documents more complex patterns. Telles and Paschel (2014) find that status *polarizes* racial identification in Brazil: income whitens, but education darkens. The apparent tension dissolves once one recognizes that their "education darkening" operates primarily within nonwhite categories—shifting individuals from *pardo* to black—rather than across the white/nonwhite boundary. In our binary outcome, this pardo-to-black shift is invisible, since both categories are coded as nonwhite. Telles and Paschel's polarization result is therefore compatible with our finding: status

simultaneously moves some individuals from *pardo* toward white (the margin we capture) and others from *pardo* toward black (the margin visible only in a multinomial framework). Both patterns reflect the broader phenomenon that higher socioeconomic status is associated with identification away from the ambiguous middle category and toward the poles of the racial distribution.

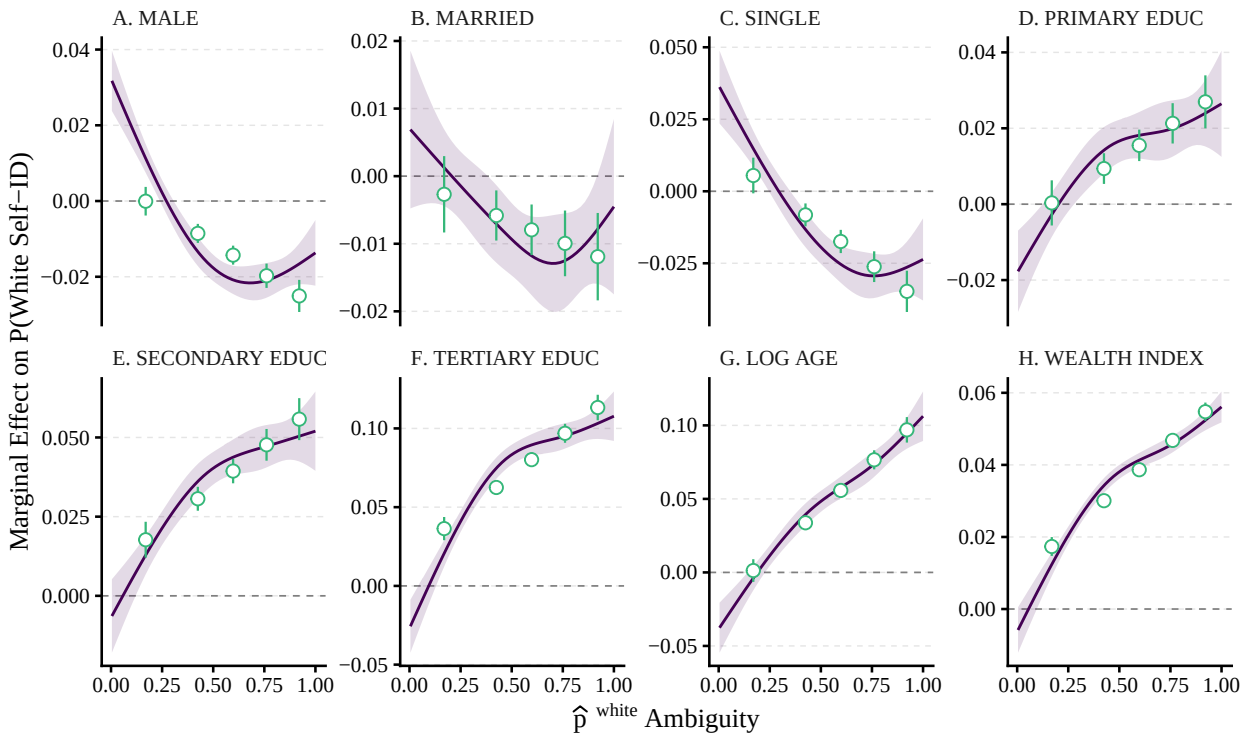

Figure 8: **Socioeconomic Effects by Racial Ambiguity.** Marginal effects of socioeconomic variables on the probability of self-identifying as white, as a function of logit-based racial ambiguity $A_i = 1 - 2|\widehat{p}_i^{\text{white}} - 0.5|$ (see Section 3.4). The curves show estimates from a cubic spline interaction model with 95% confidence bands; the points show estimates from a linear interaction model evaluated at quintile midpoints with 95% confidence intervals. Both specifications include appearance controls and municipality fixed effects. Values near 1 indicate individuals whose logit-fitted probability of white self-identification is close to 50%. Standard errors clustered at the municipality level.

Notably, the wealth index—a standardized predicted log wealth constructed by regressing log wealth on education, age, gender, marital status, occupation, and municipality fixed effects among candidates with positive asset declarations, then predicting for all candidates[8]—shows an increasing

[8]The wealth index is constructed as follows. For 2020 candidates with positive declared assets, we regress log declared assets on a full set of municipality fixed effects and individual-level demographics: sex, education attainment (three dummies), marital status (four dummies), log age, and a fixed effect for each of the TSE's individual occupation

relationship with ambiguity, with marginal effects rising from around 1.7 percentage points at low ambiguity to over 5 percentage points among the most ambiguously classified candidates. This pattern parallels education and age, suggesting that wealth reinforces social whitening most strongly precisely where appearance provides weaker classification cues. The linear model captures the overall direction well for most variables, though the spline reveals meaningful non-linearities for education that the linear specification cannot capture.

Appendix C presents robustness checks using the panel sample, white-centered ambiguity measure and the raw probability $\widehat{p}_i^{\text{white}}$. Results are broadly consistent when estimated on the panel rather than the 2020 cross-section, with appearance, education, and wealth displaying similar gradients across ambiguity bins. The panel estimates carry larger standard errors, however, which renders the gradient across ambiguity statistically insignificant for being married and attenuates the precision of the primary education gradient — patterns that are precisely estimated in the main specification. Results are also robust to replacing the $\widehat{p}^{white}$ ambiguity measure with one based on the DeepFace white-probability score (z^{white}). The direction and significance of all gradients are preserved. The main difference is that education, age, and wealth exhibit larger effects even at low ambiguity under z^{white}, producing a somewhat flatter gradient compared to the main specification. The sign reversal for being single at the lowest ambiguity bin — present in the main specification — also disappears, with the effect remaining negative throughout.

When we replace our ambiguity measure with the raw $\widehat{p}^{white}$ score — so that the x-axis runs from unambiguously non-white to unambiguously white — the previously monotone gradients become non-monotone and concave, with effects peaking among racially ambiguous candidates and declining toward both extremes. This pattern is directly informative about our ambiguity measure: low ambiguity pools together candidates at opposite ends of the appearance distribution, and the raw specification reveals that these two groups are not symmetric. For education, the effect on white self-identification is substantially larger among candidates who appear unambiguously

codes. The index is the standardized fitted value from this regression, capturing the component of wealth predicted by observable socioeconomic characteristics net of municipality-level differences. Because the index is based on predicted rather than actual assets, it assigns a value to all candidates — including those with zero or missing declared assets — as long as their occupation code appears in the estimation sample.

white than among those who appear unambiguously non-white, suggesting that education reinforces white identity most powerfully when appearance already unambiguously supports it.

Table 3 reports the coefficients on occupation group indicators from the same specifications. The reference category is "other," a heterogeneous group that includes retirees, students, homemakers, religious figures, artists and performers, media professionals, individuals living primarily on capital income, and those who did not disclose an occupation. All coefficients are therefore relative to this group. The attenuation pattern across columns is informative: without appearance or municipality controls (column 1), occupation coefficients conflate the direct association between occupation and racial self-identification with the correlation between occupation and phenotype. Adding appearance controls (column 2) increases the magnitude of the coefficient for agricultural workers (from -1.3 to -4.4 percentage points): conditional on phenotype, agricultural workers identify as nonwhite at an even higher rate than the raw association suggests, indicating that phenotypic appearance partially suppresses the occupation-identity association in the unconditional regression. Adding municipality fixed effects (column 3) further attenuates coefficients in most cases, indicating that occupational composition varies systematically across municipalities.

In the full specification with appearance and municipality controls (column 3), a clear hierarchy emerges. Professionals (+6.6 pp) and business/management occupations (+5.7 pp) are strongly associated with white self-identification, followed by commerce/sales (+3.1 pp). At the other end, manual/industrial workers (-3.6 pp) and educators (-1.8 pp) are less likely to identify as white. Agricultural workers, technicians, and public servants show effects close to zero with full controls, suggesting that their raw associations are largely explained by phenotype and geography; note that a near-zero coefficient means these groups identify as white at rates similar to the "other" reference group, not that occupation is irrelevant for them. This occupational gradient—with white-collar, higher-income occupations at the top—mirrors the education gradient and provides additional evidence for the social whitening pattern: individuals in occupations associated with higher socioeconomic status are more likely to identify as white, even conditional on appearance and municipality.

Table 3: **Occupation and Racial Self-Identification.**

	2020 Cross-Section			Panel (avg. photos)		
	(1)	(2)	(3)	(4)	(5)	(6)
Dependent Variable: Self-Reported White						
Professional	0.118***	0.069***	0.066***	0.117***	0.061***	0.062***
	(0.004)	(0.004)	(0.003)	(0.006)	(0.005)	(0.005)
Service	0.016***	0.015***	-0.012***	0.013**	0.015***	-0.013***
	(0.003)	(0.003)	(0.002)	(0.005)	(0.004)	(0.004)
Agricultural	-0.013**	-0.044***	-0.0003	-0.0004	-0.041***	0.001
	(0.006)	(0.005)	(0.002)	(0.007)	(0.005)	(0.004)
Educator	-0.065***	-0.053***	-0.018***	-0.083***	-0.067***	-0.026***
	(0.005)	(0.004)	(0.003)	(0.007)	(0.006)	(0.005)
Business/Management	0.148***	0.082***	0.057***	0.142***	0.069***	0.051***
	(0.004)	(0.003)	(0.003)	(0.006)	(0.005)	(0.004)
Public Servant	0.002	-0.014***	0.006***	-0.009*	-0.025***	0.007**
	(0.004)	(0.003)	(0.002)	(0.005)	(0.004)	(0.003)
Commerce/Sales	0.072***	0.041***	0.031***	0.068***	0.033***	0.026***
	(0.004)	(0.003)	(0.002)	(0.006)	(0.005)	(0.004)
Technician	-0.021***	-0.016***	-0.002	-0.029***	-0.023***	-0.002
	(0.005)	(0.004)	(0.003)	(0.007)	(0.006)	(0.005)
Manual/Industrial	0.023***	0.018***	-0.036***	0.012	0.014*	-0.042***
	(0.006)	(0.005)	(0.004)	(0.009)	(0.008)	(0.007)
Admin/Clerical	0.063***	0.039***	0.013**	0.062***	0.034***	0.009
	(0.007)	(0.006)	(0.005)	(0.012)	(0.010)	(0.009)
Socioeconomic Controls	✓	✓	✓	✓	✓	✓
Appearance Controls		✓	✓		✓	✓
Municipality FE			✓			✓
Observations	520,720	520,720	520,720	169,174	169,174	169,174
N Clusters	5,568	5,568	5,568	5,567	5,567	5,567
R^2	0.037	0.245	0.402	0.033	0.271	0.440
Adjusted R^2	0.037	0.245	0.396	0.033	0.271	0.421

Notes: The dependent variable is self-reported white (=1). The table shows coefficients on occupation group indicators from the same regressions as Table 2; socioeconomic controls (education, age, gender, and marital status) are included but not shown. Reference category: other (retired, students, homemakers, not disclosed). Appearance controls include five race probability categories from the DeepFace algorithm (with Latino/Hispanic as the excluded reference category). Standard errors clustered at the municipality level in parentheses. Columns (1)–(3) use the 2020 cross-section; columns (4)–(6) use the panel sample with appearance probabilities averaged across 2020 and 2024 photographs. Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Having established that socioeconomic characteristics—particularly education and occupation—shape racial self-identification conditional on appearance, and that this effect is concentrated among racially ambiguous individuals, we now turn to the historical roots of the spatial variation documented in Section 4. This connection is not coincidental: communities where historical institutions shaped the local racial context may also be communities where social whitening operates more or less intensively. The mediation analysis in Section 7 returns to this question by decomposing the slavery association into a demographic channel—local racial composition—and a residual normative channel, both of which are relevant to understanding why local context is such a strong predictor of racial self-identification for phenotypically ambiguous individuals.

6 Historical Slavery and Spatial Variation

The substantial explanatory power of municipality fixed effects documented in Section 4 suggests that local social context plays a crucial role in shaping racial self-identification. In this section, we investigate the origins of this spatial variation by examining its association with historical slavery.

6.1 Spatial Variation in Local Social Context

To capture the local social context of racial classification, we extract municipality fixed effects from a logistic regression where the dependent variable is an indicator for self-identifying as white. The explanatory variables include the appearance probabilities from the algorithm and individual socioeconomic characteristics. We interpret these fixed effects, expressed on the log-odds scale, as capturing the local social context of race—the extent to which individuals with similar phenotypic appearance and socioeconomic profiles are more or less likely to self-identify as white depending on their municipality. We use logit rather than OLS to properly account for the binary outcome; with a median of 63 candidates per municipality, the incidental parameters problem is minimal, and the logit and OLS fixed effects correlate at 0.98.[9]

[9]The incidental parameters problem arises in nonlinear models with fixed effects: as the number of groups grows, the fixed effect estimates do not converge to their true values unless the number of observations per group also grows.

Panel A of Figure 9 displays the spatial distribution of these local effects across Brazil. The Southeast and South exhibit systematically higher fixed effects than the Northeast and North, indicating a stronger local context favoring white self-identification. These regions are more developed, featuring higher levels of education and income per capita. Notably, the South experienced significant waves of European immigration in the late nineteenth century, driven by state policies aimed at promoting the "whitening" of Brazil's population (Dos Santos, 2002). We test in the following sections whether this historical social context has persisted to shape contemporary racial self-identification.

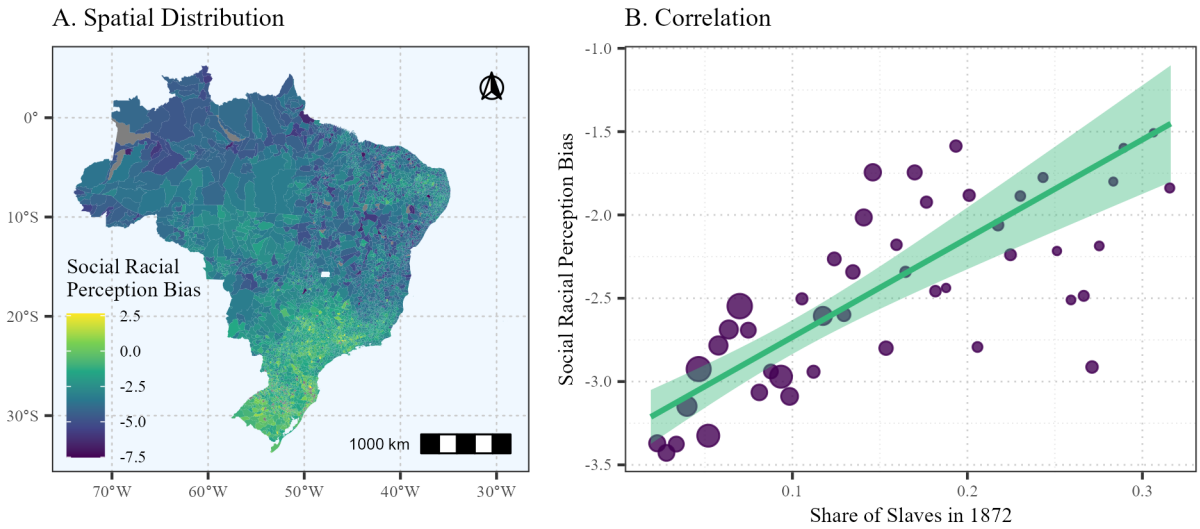

Figure 9: **Spatial variation in local social context.** Standardized municipality fixed effects estimated from a logit regression controlling for appearance probabilities and socioeconomic characteristics. Panel A displays the spatial distribution. Panel B shows the correlation with the historical intensity of slavery, measured as the share of slaves in 1872 at the minimum comparable area (AMC) level. Municipality fixed effects are first averaged within each AMC, then grouped into 50 bins by slave share. Marker size is proportional to the number of municipalities in each bin.

Panel B of Figure 9 illustrates the relationship between the local social context and the share of the slave population in 1872—the year of Brazil's first comprehensive census. To ensure consistent geographic boundaries over time, we aggregate the 1872 data to minimum comparable

Unlike in OLS, where fixed effects are estimated consistently regardless of group size, logit fixed effects are biased when groups are small. The rule of thumb is that bias becomes negligible when group size exceeds approximately 20–30 observations (Lancaster, 2000).

areas (AMCs) constructed by Reis, Pimentel, Alvarenga and Santos (2011), assigning the same slave population share to current municipalities within each area. The positive relationship is pronounced: municipalities with a social context favoring white classification tend to have had a larger share of slaves in the past. This correlation motivates a deeper investigation into the historical origins of local racial context.

6.2 The Legacy of Slavery

We regress individual-level self-reported race on the share of slaves in 1872 at the AMC level, controlling for phenotypic appearance and state fixed effects. Our identifying assumption is that, within states, conditional on geographic characteristics and spatial trends, variation in historical slavery intensity is uncorrelated with unobserved determinants of contemporary racial self-identification. This selection-on-observables strategy is standard in the historical institutions literature (Nunn and Wantchekon, 2011; Naritomi et al., 2012).

The set of plausible confounders for the relationship between historical slavery and contemporary racial self-identification is narrow. Usual confounders are associated with different levels of initial economic development that may be associated with both variables, or with geographic characteristics that may have influenced the demand and supply of slave labor and affect current development, leading to a spurious correlation between historical slavery and contemporary racial self-identification. Thus, our approach is to control for a comprehensive set of predetermined variables including geographic characteristics and initial economic development that may have led regions to have more slaves and experience an alternative development path. Post-abolition trajectories such as education provision and urbanization are more plausibly mediators than confounders, since they were shaped by the institution of slavery itself. We return to this point below.

Specifically, our estimates are conditional on coffee and sugarcane suitability from FAO GAEZ, initial economic development measured by population density from the 1872 Census, the number of pre-colonial indigenous tribes (Nimuendajú, 2017; Barsanetti and Ferreira, 2024), and proximity to ports—relevant for the slave trade—as captured by the distance-to-coast. All these variables

may have affected the demand and supply for slaves. We also add controls for terrain ruggedness, elevation, precipitation, temperature, area, and distance to rivers to account for geographic characteristics and a second-order polynomial of latitude and longitude to absorb smooth spatial trends.

Two aspects of interpretation deserve clarification. First, regarding *causality*: the controls described above address the most substantively plausible confounders, and the ambiguity heterogeneity documented below provides a strong case in favor of causal interpretation—an omitted variable that generates the observed pattern would need to not only correlate with historical slavery conditional on geography and state fixed effects, but also differentially affect individuals with ambiguous racial appearance. Second, regarding *meaning*: our slavery variable is measured at the AMC level and assigned uniformly to all candidates within each area. We interpret AMC-level variation as capturing institutional and normative environments that shaped all residents of a given area, rather than individual-level exposure differences within areas (Nunn and Wantchekon, 2011; Naritomi et al., 2012). The coefficient therefore estimates how much racial self-identification varies across areas with different historical slavery intensities, among individuals with similar appearance, holding constant state-level factors and geography.

Because slavery plausibly affected long-run education, occupation, and wealth accumulation, these socioeconomic variables are potential mediators rather than confounders; we include them as a robustness check but note that they may induce post-treatment bias (Angrist and Pischke, 2009). We condition on appearance by design, since our research question concerns how individuals with similar phenotypic appearance arrive at different racial self-identifications—isolating the channel through which slavery shapes the social meaning assigned to a given phenotype, rather than its total effect on racial composition.[10]

We build the specification progressively. Column (1) of Table 4 includes appearance probabilities and a second-order polynomial of latitude and longitude to absorb smooth spatial trends. Column (2) adds the geography controls described above. Column (3) adds individual socioeco-

[10]Appearance is itself partly a consequence of historical slavery, which shaped the racial composition of local populations over centuries.

nomic characteristics (gender, marital status, education, log age, and occupation group indicators). Columns (4)–(6) replicate the analysis using the panel sample with appearance probabilities averaged across the 2020 and 2024 photographs. Standard errors are clustered at the AMC level, the unit of variation for the slave share variable.

Table 4: **Historical Slavery and Racial Self-Identification.**

	2020 Cross-Section			Panel (avg. photos)		
	(1)	(2)	(3)	(4)	(5)	(6)
Dependent Variable: Self-Reported White						
Share of Slaves (1872)	0.287***	0.261***	0.265***	0.292***	0.250**	0.248**
	(0.105)	(0.094)	(0.098)	(0.112)	(0.100)	(0.105)
State FE	✓	✓	✓	✓	✓	✓
Geography Controls		✓	✓		✓	✓
Socioeconomic Controls			✓			✓
Observations	512,523	512,523	512,523	166,742	166,742	166,742
N Clusters	428	428	428	428	428	428
R^2	0.345	0.347	0.354	0.366	0.367	0.372
Adjusted R^2	0.345	0.347	0.353	0.365	0.367	0.372

Notes: The dependent variable is self-reported white (=1). All specifications include state fixed effects, appearance probabilities (with Latino/Hispanic as reference), and a second-order polynomial of latitude and longitude. Column (2) adds geography controls: terrain ruggedness, elevation, precipitation, temperature, log area, log distance to rivers, log distance to coast, coffee and sugarcane suitability, number of pre-colonial indigenous tribes, and 1872 population density. Column (3) adds socioeconomic controls (gender, marital status, education, log age, and occupation group indicators), which are potential mediators. Columns (4)–(6) replicate using the panel sample with averaged appearance. Standard errors clustered at the 1872 AMC level in parentheses. Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Column (1) shows that a 10 percentage point increase in the share of slaves in 1872 is associated with a 2.9 percentage point increase in the probability of self-identifying as white, conditional on appearance, state fixed effects, and spatial trends. Adding geography controls in column (2) leaves the coefficient largely unchanged (0.261), indicating that the relationship is not driven by geographic factors that may have influenced both the demand for slaves and subsequent development paths. In column (3), adding socioeconomic characteristics barely moves the coefficient (0.265): the slavery–

self-identification link does not operate through contemporary socioeconomic channels, ruling out these variables as both important confounders and important mediators. The panel sample yields similar results: the baseline association is 0.292 (column 4), attenuating modestly to 0.250 with geographic controls (column 5) and 0.248 with socioeconomic controls (column 6).

The coefficient stability across specifications is reassuring. Comparing columns (1) and (2) of Table 4, a comprehensive set of geographic controls—including crop suitability, indigenous populations, and historical population density—increases R^2 by only 0.001 while moving the coefficient from 0.287 to 0.261, a modest 9% decline. This stability suggests that selection on geographic observables is limited.

Figure 10 examines how the association between historical slavery and racial self-identification varies with racial ambiguity. We estimate interaction models—using the same specification as column (2) of Table 4—where the marginal effect of slave share is allowed to vary with logit-based ambiguity A_i, using both linear and cubic spline specifications as in the social whitening analysis. Standard errors are clustered at the 1872 AMC level. Panel A uses the cross-section sample: the linear interaction term is positive and highly significant (0.657, $p < 0.01$), indicating that the marginal effect of historical slavery on white self-identification increases with racial ambiguity; the cubic spline specification confirms this pattern, with all three spline–slavery interaction terms jointly significant at the 1% level. Panel B uses the panel sample, which averages appearance across election photographs and thus reduces measurement error in the ambiguity measure, yielding a nearly identical interaction term (0.613, $p < 0.01$) and gradient shape.

The results reveal a clear pattern: in both panels, the association between historical slavery and racial self-identification is concentrated among individuals with ambiguous phenotypic appearance. For individuals with low ambiguity—those whose appearance clearly signals either white or nonwhite—the marginal effect is close to zero and statistically insignificant. As ambiguity increases, the effect grows monotonically: at the highest ambiguity levels, a 10 percentage point increase in the 1872 slave share is associated with a 5.9 percentage point increase in the probability of identifying as white. The increasing relationship with ambiguity reinforces our earlier findings

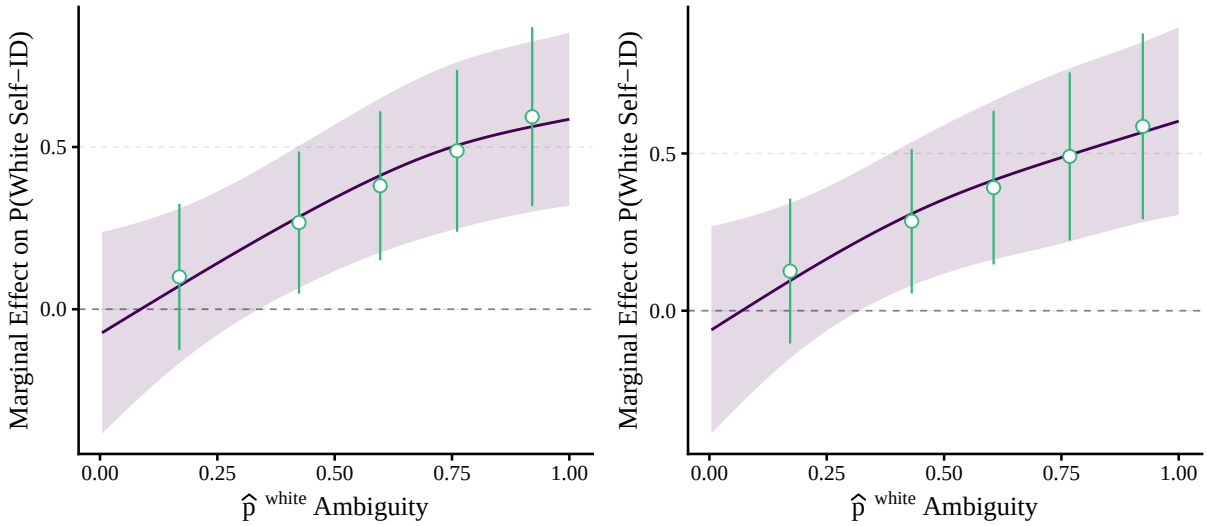

Figure 10: **Historical Slavery and Racial Self-Identification by Ambiguity.** Marginal effect of the share of slaves in 1872 on the probability of self-identifying as white, as a function of logit-based racial ambiguity $A_i = 1 - 2|\widehat{p}_i^{\text{white}} - 0.5|$ (see Section 3.4). Panel A uses the cross-section sample; Panel B uses the panel sample with averaged appearance. The curve shows estimates from a cubic spline interaction model with 90% confidence band; the points show estimates from a linear interaction model evaluated at quintile midpoints with 90% confidence intervals. All specifications include state fixed effects, appearance controls, geography controls, and a second-order polynomial of latitude and longitude. Standard errors clustered at the 1872 AMC level.

on social whitening: historical factors are most strongly associated with racial identity precisely where appearance provides the weakest signal.

In Section 7, we examine the channels through which this historical association operates using a mediation analysis and discuss the role of cultural norm transmission. Appendix D presents robustness checks using alternative ambiguity measures; the main findings are preserved. The gradient is robust to alternative appearance measures. Using white-centered ambiguity yields a significant interaction of 0.586 ($p < 0.01$); using the logit probability $\widehat{p}_i^{\text{white}}$ directly produces an interaction of 0.832 ($p < 0.01$). In both cases, the spline specifications reveal that the marginal effect levels off—and, for raw $\widehat{p}_i^{\text{white}}$, clearly declines—at the highest values, tracing an inverted-U shape peaked near the classification boundary. This is consistent with the slavery effect operating most strongly where appearance is genuinely ambiguous, attenuating for individuals whose phenotype unambiguously signals either white or nonwhite.

7 Mechanisms and Cultural Persistence

The association between historical slavery and contemporary racial self-identification documented in Section 6 raises the question of what mechanisms sustain this relationship across more than 150 years. We propose a cultural persistence story: African slavery in Brazil established social hierarchies in which nonwhite status was systematically devalued, creating normative pressure for phenotypically ambiguous individuals—those for whom racial classification is not fully determined by appearance—to identify as white. These norms were reinforced by official statistical practice: Brazil's early post-abolition censuses either excluded the race question entirely (1920) or reclassified mixed-race individuals as white (1890), effectively encoding the whitening norm into state data collection (Camargo, 2009). In areas where slavery had produced larger mixed-race populations, this institutional practice may have further entrenched the culture of white classification. These norms may have persisted through local social dynamics largely independently of the economic conditions slavery helped create. In this section, we examine the contemporary channels through which the historical association operates and show that cultural norm transmission, rather than material economic channels, is the operative mechanism.

Table 5 examines the channels through which slavery's association with contemporary self-identification operates. Both panels use the same individual-level sample. Panel A regresses five potential mediators—the contemporary white population share, the nonwhite-to-white wage gap, average years of schooling, land inequality (measured by the land Gini coefficient), and log average municipal wages—on historical slavery, conditioning on geography controls and state fixed effects. Panel B reports the direct effect of slavery on white self-identification when each mediator is added to the same baseline specification.

The results consistently rule out economic channels. Municipalities with more historical slavery have significantly higher contemporary white population shares (+0.200, $p < 0.05$, Panel A column 2), while the associations with all four economic mediators—the nonwhite-to-white wage gap, average years of schooling, the land Gini coefficient, and average wages—are all statistically indistinguishable from zero. Strikingly, historical slavery has essentially no association with

Table 5: **Mediation Analysis: Channels of Slavery's Association.**

	% White	Wage Gap	School.	Land Gini	Wage
	(1)	(2)	(3)	(4)	(5)
Panel A - Dependent Variable: Mediator					
Share of Slaves (1872)	0.200**	-0.123	-0.535	-0.037	-0.003
	(0.081)	(0.078)	(0.771)	(0.072)	(0.260)
Observations	512,523	512,523	512,523	510,585	512,523
N Clusters	428	428	428	428	428
R^2	0.841	0.264	0.450	0.344	0.617
Panel B - Dependent Variable: Self-Reported White					
Share of Slaves (1872)	0.124**	0.257***	0.262***	0.265***	0.262***
	(0.062)	(0.095)	(0.094)	(0.095)	(0.099)
Mediator	0.698***	-0.033	0.003	-0.017	0.056***
	(0.032)	(0.026)	(0.002)	(0.032)	(0.008)
Indirect Effect	0.139	0.004	-0.002	0.0006	-0.0002
Sobel Test	2.44	0.989	-0.612	0.370	-0.010
Observations	512,523	512,523	512,523	510,585	512,523
N Clusters	428	428	428	428	428
R^2	0.361	0.347	0.347	0.347	0.348

Notes: Both panels use the same individual-level sample as column (2) of Table 4. Panel A reports OLS regressions with the column-header variable as the dependent variable; the five mediators are the 2022 white population share, the nonwhite-to-white wage gap, average years of schooling, the land Gini coefficient, and log average municipal wages. Because these mediators are municipality-level variables, the Panel A coefficients are equivalent to municipality-level estimates; using the individual-level dataset ensures consistency with Panel B for the Sobel standard errors. Panel B reports OLS regressions with white self-identification as the dependent variable, adding the column-header mediator to the baseline specification (appearance probabilities, geography controls, latitude-longitude polynomial, and state fixed effects). "Mediator" reports the coefficient on the mediator variable itself. Column (1) of Panel B is the total effect without any mediator. The Sobel test statistic is $z = \hat{\alpha}\hat{\beta}_M / \sqrt{\hat{\beta}_M^2 \widehat{\text{se}}_\alpha^2 + \hat{\alpha}^2 \widehat{\text{se}}_{\beta_M}^2}$, where $\hat{\alpha}$ is the column-specific Panel A coefficient (the effect of the 1872 slave share on the column-header mediator) and $\hat{\beta}_M$ is the mediator's coefficient in Panel B; p-values are from the standard normal. Standard errors clustered at the 1872 AMC level. Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

average wages today (−0.003, n.s.), indicating that slavery's long-run effect on racial identity does not operate through any persistent impact on municipal economic development. Panel B confirms

the pattern: conditioning on any of the four economic mediators leaves the slavery coefficient essentially unchanged (0.257–0.265), and none of the corresponding Sobel tests rejects the null of zero indirect effect ($p \geq 0.32$ in all cases). This provides no support for stories in which individuals lighten their racial identity in response to wage-based incentives, historically rooted inequality, or aggregate development levels.

Average wages are themselves positively associated with white self-identification in Panel B (coefficient = 0.056, $p < 0.01$)—consistent with the concern that confounders related to slavery intensity and divergent development paths could bias the results. But this association is independent of slavery's effect: since slavery does not predict average wages, the wage-whiteness correlation does not translate into an indirect path from slavery to racial identity (Sobel $p = 0.992$).

The white population share stands apart: controlling for it reduces the slavery coefficient from 0.261 to 0.124, and the Sobel test rejects the null of zero indirect effect (indirect = 0.139, $p = 0.015$). Yet this result cannot be interpreted as evidence of causal mediation, because two distinct mechanisms could generate it. First, a *reference-group* mechanism: candidates in municipalities with more white residents may self-classify as white because their reference group is whiter, shifting the local threshold for who "counts" as white. Relatedly, since our sample is composed of political candidates, they may be particularly sensitive to the local racial composition of their electorate, and thus more likely to identify as white in whiter municipalities to appeal to voters. Janusz (2021) indeed finds that candidates change their racial self-identification across elections in response to changes in the racial composition of their electorate.

Second, a *parallel-consequence* mechanism: because the white population share in the 2022 Census is itself, self-reported both the contemporary white population share and candidate self-identification may be parallel outcomes of the same latent cultural norm—the historical pressure toward white identification—rather than being causally linked. These two mechanisms cannot be disentangled without phenotypic data on the general population (cf. Chor et al., 2019, who show that local racial composition modifies how ancestry maps to self-declared race). It is therefore impossible to determine whether high-slavery municipalities have more white-identifying residents

because their phenotypic composition differs, or because residents face the same social pressure toward white self-identification that we document for political candidates. If the latter, white share and candidate white self-identification are parallel outcomes of a common latent norm, and the Sobel result simply reflects that the norm operates on all residents, not just candidates. We interpret the finding descriptively: slavery is associated with higher rates of white self-identification among both census residents and political candidates, a pattern consistent with area-level social norms, but one that cannot be decomposed into compositional versus normative channels without phenotypic data on the general population.

Regardless of the mechanism connecting share of whites and slavery, the residual effect of slavery on candidate white self-identification is still significant (0.124). This is consistent with a cultural persistence story. Areas with more intensive historical slavery developed social hierarchies in which nonwhite status was equated with servitude and low social standing. These hierarchies likely generated a persistent normative environment in which phenotypically ambiguous individuals—those whose appearance does not unambiguously determine their racial category—faced sustained pressure to identify as white (see Telles, 2004).

The primacy of cultural over economic channels echoes Nunn and Wantchekon's (2011) finding that the African slave trade's persistent effects on mistrust operate primarily through internal cultural norms—at least twice as large as the external institutional channel—rather than through persistent economic structures. High-slavery areas today have larger white populations—consistent with the same social pressures operating across the broader population—and ruling out wage-based incentives, educational gradients, land inequality, and aggregate development as operative channels narrows the explanation toward a cultural transmission that persists largely independently of contemporary material circumstances.

The ambiguity gradient provides the sharpest evidence for this interpretation. Figure 11 examines whether the slavery–self-identification association persists after absorbing the demographic channel, by conditioning on municipality demographics (white share, wage gap, and years of schooling). Panel A uses the cross-section sample: for low-ambiguity individuals (bottom quin-

tile), adding these controls drives the estimated effect close to zero (−0.102, n.s.), so their slavery association is fully accounted for by the compositional channel; for high-ambiguity individuals (top quintile), by contrast, a large positive effect of 0.39 remains—roughly two-thirds of their baseline estimate. Panel B confirms this pattern in the panel sample. In both panels, the ambiguity gradient survives absorption of the compositional channel and becomes more pronounced: the slavery association is concentrated precisely where racial classification is most discretionary.

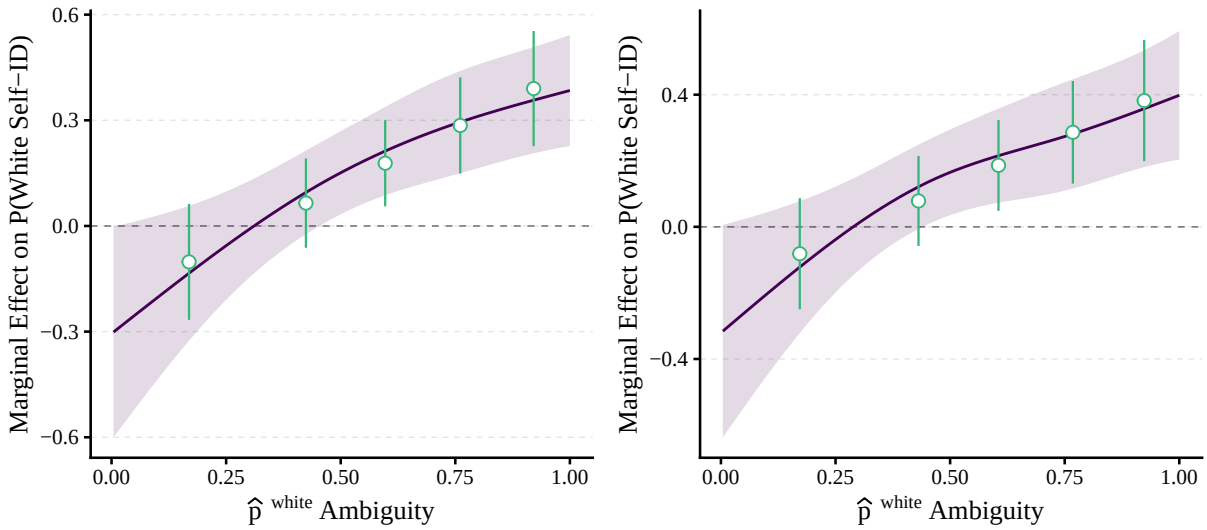

Figure 11: **Slavery Association by Ambiguity, Controlling for Municipality Demographics.** Marginal effect of the historical slave share (1872) on the probability of self-identifying as white, as a function of logit-based racial ambiguity A_i, after adding municipality-level demographic controls (2022 white population share, nonwhite-to-white wage gap, and average years of schooling) to the baseline specification. Panel A uses the cross-section sample; Panel B uses the panel sample with averaged appearance. The spline estimate (shaded band, 90% CI) and the linear interaction at quintile midpoints (dots with 90% CI) both show that the positive ambiguity gradient survives absorption of the demographic channel. Standard errors clustered at the 1872 AMC level.

This pattern is consistent with cultural norm transmission that operates most strongly when appearance permits flexibility. Individuals whose phenotype unambiguously determines their racial category have limited latitude to respond to social pressure; the whitening norm operates mainly for those whose appearance leaves classification open. The persistence of a large residual effect among high-ambiguity individuals after conditioning on demographics suggests the mechanism goes beyond a simple reference-group effect: it reflects historically rooted normative expectations

that continue to influence racial self-identification for phenotypically mixed individuals regardless of their immediate demographic environment. Areas that relied more heavily on enslaved labor developed a cultural context in which mixed-race individuals faced—and in many cases still face—systematic pressure to align their self-reported identity with whiteness, a pattern that has outlasted both slavery itself and the economic structures it produced.

Appendix D presents robustness checks for the slavery heterogeneity patterns using the white-centered ambiguity measure and the panel sample. The main findings are preserved across these alternatives.

8 Conclusion

This paper provides new evidence on the relationship between phenotypic appearance, socioeconomic characteristics, and social context in shaping racial identity in Brazil. Using machine learning-based phenotypic classification alongside self-reported race from a large sample of political candidates, we show that while physical appearance strongly predicts racial self-identification, local social context—captured through municipality fixed effects—plays an even larger role, particularly among individuals with racially ambiguous phenotypes. Socioeconomic factors such as education and wealth are also associated with white self-identification, with these "social whitening" effects concentrated among the phenotypically ambiguous.

We further show that local social context has deep historical roots. Municipalities with higher slave population shares in 1872 display stronger norms favoring whiteness today, and this association is concentrated among individuals whose appearance does not clearly signal a racial category. A mediation analysis reveals that contemporary racial composition—itself potentially a parallel outcome of the same latent norm—accounts for approximately 54% of this association, while economic channels carry none of it. These patterns are consistent with cultural persistence: the social hierarchies established under slavery created normative pressure for mixed-race individuals to identify as white—pressure that has outlasted both slavery itself and the economic structures it

produced.

Our analysis uses a cross-sectional design with a self-selected sample of political candidates, which limits causal claims about social whitening and may not generalize to the broader population. The slavery association rests on a selection-on-observables strategy. These caveats notwithstanding, the findings underscore that racial classification in Brazil cannot be understood through appearance alone—it is fundamentally shaped by social context with roots in the country's history of slavery.

Appendix A Dataset Construction

This appendix documents the construction of the analysis dataset. The final dataset is a candidate-level panel covering the 2020 and 2024 Brazilian municipal elections, organized in long form with a `year` indicator. Four separate data-building steps produce the required inputs, which are then merged at the candidate–municipality level.

Candidate and electoral data. Individual-level data on candidates come from the *Tribunal Superior Eleitoral* (TSE). For each election year, we use the official candidacy files (`consulta_cand`), which contain self-declared race, gender, date of birth, marital status, educational attainment, party affiliation, and campaign spending limits (available only for 2020). Candidates are restricted to first-round participants in regular municipal elections. For 2020 we additionally require an active candidacy status (`CD_SITUACAO_CANDIDATURA = 12`); for 2024 we exclude records flagged with an invalid election outcome status (`CD_SIT_TOT_TURNO` $\neq$ `-1`). A small number of duplicate records (one in 2020, two in 2024) are manually removed.

Wealth is computed from the TSE asset declaration files (`bem_candidato`), which list the declared value of each asset held by a candidate. We sum all reported asset values per candidate to obtain a single wealth measure. Because reported values use a comma as the decimal separator, they are converted to numeric format before aggregation.

The dependent variable is a binary indicator equal to one if the candidate's self-reported race is *branca* (white). Candidate age is obtained directly from the TSE field `NR_IDADE_DATA_POSSE` in 2020; for 2024, where this variable is unavailable, age is computed from the date of birth. Ages above 100 are set to missing. Campaign spending values equal to or below zero (2020 only) are also set to missing.

Education is recoded into binary indicators for each attainment level (literate, incomplete primary, primary, incomplete secondary, secondary, incomplete tertiary, and tertiary), with *illiterate* as the omitted category. We further aggregate these into three broader categories: primary attainment (completed primary or incomplete secondary), secondary attainment (completed secondary

or incomplete tertiary), and tertiary attainment (completed tertiary). Marital status is recoded into indicators for single, married, and formerly married (separated, divorced, or widowed).

Occupation. Candidates report their occupation to the TSE using a detailed classification system (CD_OCUPACAO) with over 100 categories. We aggregate these into 10 broad sectoral groups: professional (lawyers, doctors, engineers, and similar), educator (teachers and professors), technician (technical occupations), business and management (entrepreneurs, directors, and managers), public servant (elected officials and civil servants), administrative and clerical (office and administrative workers), commerce and sales (merchants, salespeople, and brokers), service (security, transportation, domestic, and personal services), agricultural (farmers, ranchers, and fishers), and manual/industrial (factory workers, construction workers, and artisans). The residual category, *other*, includes retirees, students, religious figures, artists and performers, media professionals (journalists, broadcasters), individuals living primarily on capital income, and those who did not disclose an occupation. All regressions include occupation group indicators with *other* as the reference category.

Phenotypic appearance. Candidate photographs are processed through the DeepFace pipeline using two face-detection algorithms: RetinaFace (our baseline) and OpenCV (used for robustness). Each algorithm produces a six-dimensional probability vector over the categories *Asian*, *Indian*, *Black*, *White*, *Middle Eastern*, and *Latino/Hispanic*. The baseline RetinaFace probabilities are stored without suffix; OpenCV probabilities carry a _cv suffix. All probabilities are rescaled from percentages to the unit interval. The candidate-level dataset is formed by an inner join with the RetinaFace output (so a candidate must have a successful face detection to enter the sample) and a left join with the OpenCV output (allowing for missing values when OpenCV fails to detect a face).

Census data. Municipality-level demographic characteristics are drawn from IBGE sources. From the 2022 Population Census (accessed via SIDRA), we obtain the total population, white population, literate population, and male population for each municipality, from which we compute

the white, male, and literate population shares. Average wages by race are also obtained from SIDRA and used to construct the nonwhite-to-white and black-to-white wage gaps. Missing wage values are replaced with zero before computing the ratios.

Historical data on slavery come from the 1872 Brazilian Census. Because municipal boundaries have changed substantially since 1872, we use minimum comparable areas (AMCs) to map 1872 data to modern municipalities. The AMC crosswalk assigns each current municipality a unique 1872 comparable area code, and the share of enslaved individuals in the total population is computed at the AMC level.

Geographic controls. Municipality-level geographic characteristics are extracted from raster datasets using zonal statistics. Median values of terrain ruggedness, elevation, precipitation, and temperature are computed for each municipality polygon (2010 boundaries) using exact raster extraction. Temperature values from the source raster are divided by 10 to convert to degrees Celsius. Distance measures—to the nearest river, to the coastline, and to the Tordesillas line (meridian at $-48.7°$ longitude)—are computed from municipal seat coordinates projected onto the South America Equidistant Conic coordinate system (ESRI:102032). Municipality area is computed under the South America Albers Equal Area projection (ESRI:102033) and expressed in square kilometers.

Panel linkage. Candidates are linked across the 2020 and 2024 elections using their voter registration number (`NR_TITULO_ELEITORAL_CANDIDATO`). Duplicate registration numbers within a given year are removed before linking. For candidates observed in both years, we compute: (i) the average phenotypic appearance probabilities across the two photographs (for both RetinaFace and OpenCV), (ii) a binary indicator for switching from nonwhite to white self-identification, and (iii) a binary indicator for switching from white to nonwhite self-identification. These cross-year variables are set to missing for candidates who ran in only one election. In all panel regressions, the dependent variable is the 2020 self-reported race; the panel is used exclusively to reduce measurement error in appearance by averaging photographs across years, not to construct a longitudinal

outcome measure.

Summary statistics. Table A.1 reports descriptive statistics for the 2020 cross-section. The sample contains approximately 527,000 candidates, of whom 48.3% self-identify as white. The mean probability of being classified as white by DeepFace is 0.343, with substantial variation (SD = 0.285). Candidates are predominantly male (66.5%), and the modal education level is secondary (42.7%). The largest occupation groups are *other* (28.4%, a residual group including retirees, students, religious figures, artists, media professionals, and undisclosed occupations), public servants (14.6%), and service workers (13.0%). Table A.2 reports corresponding statistics for the panel sample of approximately 169,000 candidates who ran in both elections. The panel is similar in composition, though candidates are more likely to be male (74.9%) and to work in the public sector (21.9%), consistent with incumbency and repeat candidacy.

Table A.1: **Summary statistics — 2020**

Variable	N	Mean	SD	Min	P25	Median	P75	Max
z^{White}	520739	0.34	0.28	0.00	0.13	0.25	0.50	1.00
z^{Asian}	520739	0.10	0.16	0.00	0.00	0.03	0.12	1.00
z^{Black}	520739	0.05	0.15	0.00	0.00	0.01	0.04	1.00
z^{Indian}	520739	0.07	0.08	0.00	0.01	0.05	0.10	1.00
$z^{\text{Middle Eastern}}$	520739	0.16	0.15	0.00	0.06	0.13	0.22	1.00
$z^{\text{Latino/Hispanic}}$	520739	0.27	0.16	0.00	0.16	0.27	0.37	0.97
Self-Reported White	526965	0.48	0.50	0.00	0.00	0.00	1.00	1.00
Male	526965	0.67	0.47	0.00	0.00	1.00	1.00	1.00
Married	526965	0.52	0.50	0.00	0.00	1.00	1.00	1.00
Single	526965	0.37	0.48	0.00	0.00	0.00	1.00	1.00
Formerly Married	526965	0.11	0.32	0.00	0.00	0.00	0.00	1.00
Primary Education	526965	0.17	0.38	0.00	0.00	0.00	0.00	1.00
Secondary Education	526965	0.43	0.49	0.00	0.00	0.00	1.00	1.00
Tertiary Education	526965	0.25	0.43	0.00	0.00	0.00	0.00	1.00
Age	526958	45.55	11.52	18.00	37.00	45.00	54.00	99.00
Professional	526965	0.06	0.24	0.00	0.00	0.00	0.00	1.00
Educator	526965	0.05	0.22	0.00	0.00	0.00	0.00	1.00
Technician	526965	0.04	0.19	0.00	0.00	0.00	0.00	1.00
Business/Management	526965	0.07	0.25	0.00	0.00	0.00	0.00	1.00
Public Servant	526965	0.15	0.35	0.00	0.00	0.00	0.00	1.00
Admin/Clerical	526965	0.01	0.12	0.00	0.00	0.00	0.00	1.00
Commerce/Sales	526965	0.09	0.28	0.00	0.00	0.00	0.00	1.00
Service	526965	0.13	0.34	0.00	0.00	0.00	0.00	1.00
Agricultural	526965	0.10	0.30	0.00	0.00	0.00	0.00	1.00
Manual/Industrial	526965	0.03	0.16	0.00	0.00	0.00	0.00	1.00
Other	526965	0.28	0.45	0.00	0.00	0.00	1.00	1.00

Table A.2: **Summary statistics — Panel**

Variable	N	Mean	SD	Min	P25	Median	P75	Max
z^{White} (Avg.)	169177	0.35	0.27	0.00	0.15	0.27	0.52	1.00
z^{Asian} (Avg.)	169177	0.09	0.14	0.00	0.01	0.04	0.12	1.00
z^{Black} (Avg.)	169177	0.05	0.14	0.00	0.00	0.01	0.04	1.00
z^{Indian} (Avg.)	169177	0.07	0.06	0.00	0.02	0.05	0.10	0.98
$z^{\text{Middle Eastern}}$ (Avg.)	169177	0.17	0.14	0.00	0.07	0.14	0.22	1.00
$z^{\text{Latino/Hispanic}}$ (Avg.)	169177	0.27	0.14	0.00	0.17	0.27	0.36	0.93
Self-Reported White	169216	0.49	0.50	0.00	0.00	0.00	1.00	1.00
Male	169216	0.75	0.43	0.00	0.00	1.00	1.00	1.00
Married	169216	0.57	0.50	0.00	0.00	1.00	1.00	1.00
Single	169216	0.33	0.47	0.00	0.00	0.00	1.00	1.00
Formerly Married	169216	0.10	0.30	0.00	0.00	0.00	0.00	1.00
Primary Education	169216	0.16	0.37	0.00	0.00	0.00	0.00	1.00
Secondary Education	169216	0.42	0.49	0.00	0.00	0.00	1.00	1.00
Tertiary Education	169216	0.27	0.45	0.00	0.00	0.00	1.00	1.00
Age	169213	45.60	10.46	18.00	38.00	45.00	53.00	90.00
Professional	169216	0.06	0.25	0.00	0.00	0.00	0.00	1.00
Educator	169216	0.05	0.22	0.00	0.00	0.00	0.00	1.00
Technician	169216	0.04	0.20	0.00	0.00	0.00	0.00	1.00
Business/Management	169216	0.07	0.26	0.00	0.00	0.00	0.00	1.00
Public Servant	169216	0.22	0.41	0.00	0.00	0.00	0.00	1.00
Admin/Clerical	169216	0.01	0.11	0.00	0.00	0.00	0.00	1.00
Commerce/Sales	169216	0.08	0.28	0.00	0.00	0.00	0.00	1.00
Service	169216	0.10	0.30	0.00	0.00	0.00	0.00	1.00
Agricultural	169216	0.11	0.31	0.00	0.00	0.00	0.00	1.00
Manual/Industrial	169216	0.02	0.15	0.00	0.00	0.00	0.00	1.00
Other	169216	0.23	0.42	0.00	0.00	0.00	0.00	1.00

Appendix B Extensions: Shapley Decomposition

This appendix examines the robustness of the Shapley-Owen decomposition analysis presented in Section 4 to alternative samples and ambiguity measures. Figure B.1 presents the decomposition across ambiguity quintiles for three specifications: the panel sample with averaged photographs (Panel A), the white-centered ambiguity measure based on the raw DeepFace white probability (Panel B), and the raw logit-based probability $\widehat{p}_i^{\text{white}}$ without the ambiguity transformation (Panel C). In the main text, Figure 7 uses the logit-based ambiguity measure $A_i = 1 - 2|\widehat{p}_i^{\text{white}} - 0.5|$ on the 2020 cross-section.

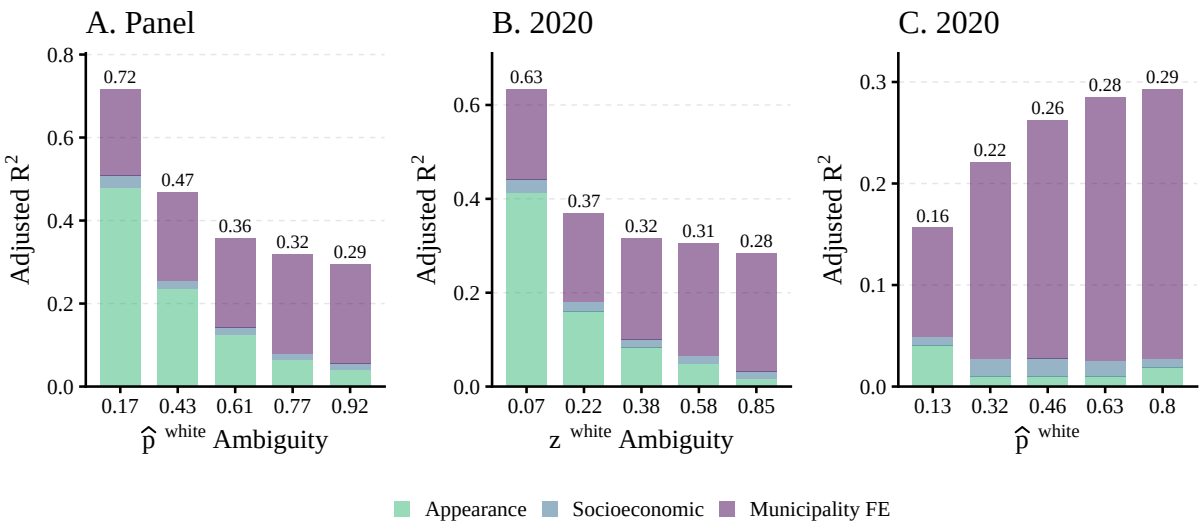

Figure B.1: **Shapley-Owen decomposition—robustness checks.** Each panel shows Shapley values in adjusted R^2 units, computed within quintiles of the indicated measure. Panel A replicates the main ambiguity analysis from Figure 7 on the panel sample using averaged photographs. Panel B uses the white-centered ambiguity measure $A_i^{wc} = 1 - 2|z_i^{\text{white}} - 0.5|$. Panel C plots the decomposition against the raw logit-based probability $\widehat{p}_i^{\text{white}}$, separating clearly nonwhite (left) from clearly white (right) individuals.

Panel A replicates the main analysis on the panel sample, where each candidate's appearance probabilities are averaged across their 2020 and 2024 photographs ($N \approx 167{,}000$). The pattern closely mirrors the main text: appearance contributions decline from 0.479 (66.9%) in the lowest ambiguity quintile to 0.040 (13.5%) in the highest, while municipality fixed effects rise from 0.209 (29.1%) to 0.239 (81.0%). The overall R^2 gradient—from 0.72 to 0.30—is also comparable to the

cross-sectional decline from 0.70 to 0.27. The slightly higher appearance share at high ambiguity in the panel (13.5% vs. 4.0% in the main text) likely reflects reduced measurement error from averaging two photographs, which attenuates the collapse in appearance's predictive power.

Panel B uses the white-centered ambiguity measure $A_i^{wc} = 1 - 2|z_i^{\text{white}} - 0.5|$, which relies on the raw DeepFace white probability directly rather than the logit-fitted value. The results are nearly identical to the main text: appearance contributions decline from 0.412 (65.0%) to 0.017 (5.8%) across quintiles, and municipality fixed effects increase from 30.5% to 88.9% of explained variation. Total R^2 falls from 0.63 to 0.28. The close correspondence confirms that the key finding—social context becoming dominant at high ambiguity—is not an artifact of the logit-based measure.

Panel C plots the decomposition directly against the logit-based probability $\widehat{p}_i^{\text{white}}$, without the ambiguity transformation. Unlike the ambiguity measures, which are symmetric around 0.5, $\widehat{p}_i^{\text{white}}$ separates clearly nonwhite individuals (low values) from clearly white individuals (high values). At low $\widehat{p}_i^{\text{white}}$ (Q1, median $\approx$ 0.13), total R^2 is only 0.16, with appearance accounting for 26.1% and municipality fixed effects for 68.8%. As $\widehat{p}_i^{\text{white}}$ increases, appearance's share drops sharply—to 4.6% in Q2 and remains between 3.5% and 6.4% thereafter—while municipality fixed effects dominate at 88–91%. This asymmetry is informative: clearly nonwhite individuals have low overall R^2 (their near-universal nonwhite self-identification leaves little variation to explain), and what variation exists is partly attributable to appearance. Among individuals with moderate to high $\widehat{p}_i^{\text{white}}$, municipality fixed effects account for the vast majority of explained variation, reinforcing the conclusion that where one lives matters most for racial self-identification when appearance does not decisively predict the outcome.

Two-group decomposition. As a further check, we omit appearance probabilities entirely and decompose the adjusted R^2 into only two groups: socioeconomic characteristics and municipality fixed effects. In the 2020 cross-section, the total adjusted R^2 without appearance is 0.291—substantially lower than the 0.396 obtained with all three groups in the main specification. Of this total, municipality fixed effects account for 0.263 (90.5%) and socioeconomic characteristics for

only 0.028 (9.5%). This confirms that local social context is by far the dominant non-appearance determinant of racial self-identification: even when all phenotypic information is excluded, where a candidate lives explains nine times more variation than individual socioeconomic characteristics.

Logit specification. The main text reports results from linear probability models, where coefficients represent marginal effects and R^2 has a direct interpretation as the share of variance explained. Here we verify that results are robust to estimating logit models, which correctly account for the binary nature of the dependent variable.

Table B.1 replicates Table 1 using logit estimation. Coefficients are reported on the log-odds scale. The pattern of results closely mirrors the OLS findings. The probability of being algorithmically classified as white remains a strong positive predictor of self-identifying as white, with coefficients ranging from 2.52 to 3.79 depending on the specification. *Middle Eastern* appearance is also positively associated with white self-identification, while black and *Indian* appearances show strong negative associations. The relative magnitudes across appearance categories are preserved, and the attenuation of coefficients when adding controls follows the same pattern as in the linear models.

We also replicate the Shapley-Owen decomposition using McFadden's pseudo R^2, defined as $R^2_{McF} = 1 - \ln \mathcal{L}_{full} / \ln \mathcal{L}_{null}$, where $\mathcal{L}_{full}$ is the likelihood of the estimated model and $\mathcal{L}_{null}$ is the likelihood of the intercept-only model. Unlike OLS R^2, pseudo R^2 measures the proportional improvement in log-likelihood rather than the share of variance explained. Values are typically lower, and the decomposition should be interpreted as contributions to model fit. Despite this difference, the Shapley-Owen decomposition remains valid: marginal contributions to pseudo R^2 are well-defined, and the Shapley values sum to the total.

The aggregate pseudo R^2 decomposition yields: appearance (0.152, 45.1%), socioeconomic (0.017, 5%), and municipality fixed effects (0.168, 49.9%), for a total of 0.337. Compared to the OLS decomposition—where appearance accounted for 42% and municipality fixed effects for 53%—the logit specification still attributes a larger share to municipality fixed effects. This

Table B.1: **Determinants of Racial Self-Identification (Logit).**

	2020 Cross-Section			Panel (avg. photos)		
	(1)	(2)	(3)	(4)	(5)	(6)
Dependent Variable: Self-Reported White						
z^{White}	3.12***	3.00***	2.52***	3.79***	3.64***	3.08***
	(0.034)	(0.034)	(0.032)	(0.058)	(0.058)	(0.065)
z^{Asian}	0.670***	0.584***	0.607***	1.28***	1.14***	1.08***
	(0.034)	(0.034)	(0.035)	(0.062)	(0.062)	(0.076)
z^{Black}	-4.78***	-4.73***	-5.49***	-5.04***	-5.06***	-6.31***
	(0.144)	(0.141)	(0.146)	(0.247)	(0.246)	(0.277)
z^{Indian}	-3.19***	-3.23***	-3.19***	-4.36***	-4.48***	-4.96***
	(0.101)	(0.101)	(0.101)	(0.209)	(0.212)	(0.237)
$z^{\text{Middle Eastern}}$	1.65***	1.58***	1.48***	2.18***	2.08***	1.82***
	(0.034)	(0.034)	(0.035)	(0.062)	(0.063)	(0.073)
Socioeconomic Controls		✓	✓		✓	✓
Municipality FE			✓			✓
Observations	520,733	520,733	515,848	169,175	169,175	162,010
N Clusters	5,568	5,568	5,428	5,567	5,567	5,114
Pseudo R^2	0.191	0.204	0.353	0.216	0.228	0.378
Adjusted Pseudo R^2	0.191	0.203	0.338	0.216	0.227	0.332
Log-Likelihood	-291,929.7	-287,239.4	-231,175.2	-91,857.1	-90,560.9	-69,846.4

Notes: The dependent variable is self-reported white (=1). Coefficients are on the log-odds scale. All appearance probabilities are from the DeepFace algorithm; Latino/Hispanic is the reference category. Socioeconomic controls include gender, marital status, educational attainment, log age, and occupation group indicators. Standard errors clustered at the municipality level in parentheses. Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

difference reflects how the two measures weight observations: pseudo R^2 is more sensitive to correct predictions in the tails of the probability distribution, where municipality-level variation may be more informative.

Figure B.2 replicates the heterogeneity analysis by logit-based ambiguity A_i. The qualitative pattern is similar: appearance contributions decline with ambiguity (from 78.0% in the least ambiguous quintile to 6.2% in the most ambiguous), and total pseudo R^2 falls substantially (from 0.57 to 0.15). Like the OLS results, municipality fixed effects increase monotonically; thus the core finding is preserved: appearance loses explanatory power as racial ambiguity increases, and social context accounts for a larger share of model fit among individuals with ambiguous appearances.

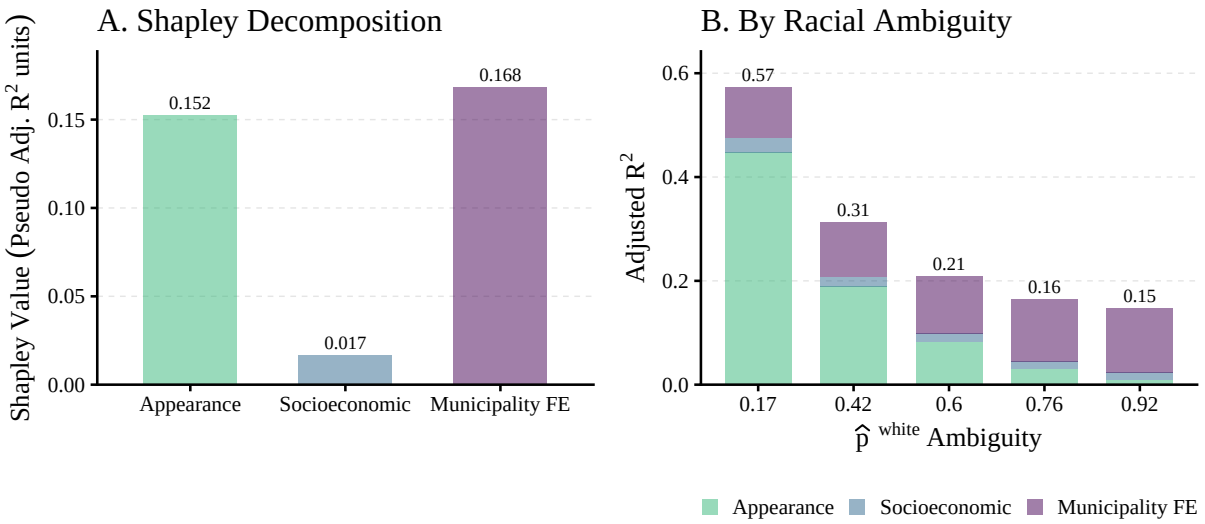

Figure B.2: **Shapley-Owen decomposition of pseudo R^2 by racial ambiguity (logit).** Each stacked bar shows Shapley values in pseudo R^2 units, computed within quintiles of logit-based ambiguity A_i (see Section 3.4). Total bar height equals the full-model McFadden pseudo R^2 for each quintile.

Appendix C Extensions: Social Whitening

This appendix examines the robustness of the socioeconomic heterogeneity analysis presented in Figure 8. In the main text, the marginal effects of socioeconomic characteristics on white self-identification are plotted against the logit-based ambiguity measure $A_i = 1 - 2|\widehat{p}_i^{\text{white}} - 0.5|$ for the 2020 cross-section. Here we replicate the analysis using the panel sample (Panel A), the white-centered ambiguity measure $A_i^{wc} = 1 - 2|z_i^{\text{white}} - 0.5|$ (Panel B), and the raw logit-based probability $\widehat{p}_i^{\text{white}}$ (Panel C). In each case, figures report both the linear interaction estimates evaluated at quintile midpoints (points with error bars) and cubic spline estimates with 95% confidence bands, following the specifications in Equations (3) and (4).

Panel sample. Figure C.1 replicates the analysis on the panel sample, where appearance probabilities are averaged across 2020 and 2024 photographs ($N \approx 167{,}000$). The education gradient is preserved: tertiary education increases white self-identification by 3.5 percentage points at low ambiguity ($A \approx 0.17$), rising to 10.2 percentage points at high ambiguity ($A \approx 0.92$), compared to 3.6–11.3 percentage points in the main specification. Secondary education and log age display similar increasing gradients. The wealth index is also robust, rising from 2.0 to 5.5 percentage points—nearly identical to the main text (1.7 to 5.5 pp). The panel estimates carry larger standard errors, however, which renders the gradient for married statistically insignificant across most of the ambiguity distribution and attenuates the precision of the primary education gradient. The male coefficient is positive at low ambiguity (+0.9 pp) and turns negative at high ambiguity (−1.6 pp), consistent with the main specification where the male effect crosses zero.

White-centered ambiguity. Figure C.2 uses the white-centered measure $A_i^{wc} = 1 - 2|z_i^{\text{white}} - 0.5|$, which relies on the raw DeepFace white probability rather than the logit-fitted value. The direction and significance of all gradients are preserved. Tertiary education increases from 5.9 to 10.5 percentage points across the ambiguity distribution, and log age from 2.6 to 9.3 percentage points. The wealth index rises from 2.9 to 5.1 percentage points. The main difference relative to the

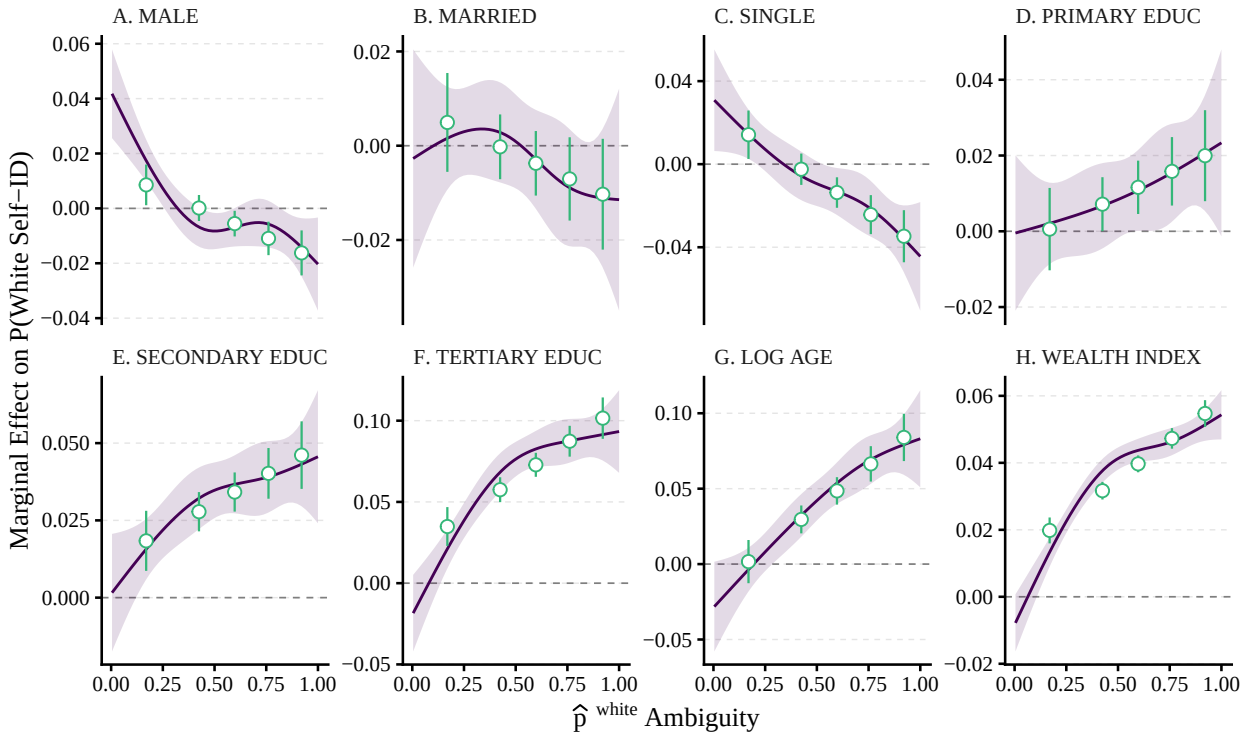

Figure C.1: **Socioeconomic effects by racial ambiguity—panel sample.** Marginal effects of socioeconomic variables on the probability of self-identifying as white, as a function of logit-based racial ambiguity $A_i = 1 - 2|\widehat{p}_i^{\text{white}} - 0.5|$ (see Section 3.4). Appearance probabilities are averaged across 2020 and 2024 photographs. The curves show estimates from a cubic spline interaction model with 95% confidence bands; the points show estimates from a linear interaction model evaluated at quintile midpoints with 95% confidence intervals. Both specifications include appearance controls and municipality fixed effects. Standard errors clustered at the municipality level.

main specification is that education, age, and wealth exhibit larger effects even at low ambiguity under z^{white}, producing a somewhat flatter gradient. For example, tertiary education at the lowest quintile is 5.9 percentage points under z^{white} versus 3.6 under $\widehat{p}^{\text{white}}$. This reflects the fact that the z^{white}-based measure classifies some individuals as "unambiguous" who the logit assigns moderate ambiguity to, diluting the low-ambiguity bin with moderately ambiguous individuals. The sign reversal for being single at the lowest ambiguity bin—present in the main specification—also disappears, with the effect remaining negative throughout (−0.7 to −2.8 pp).

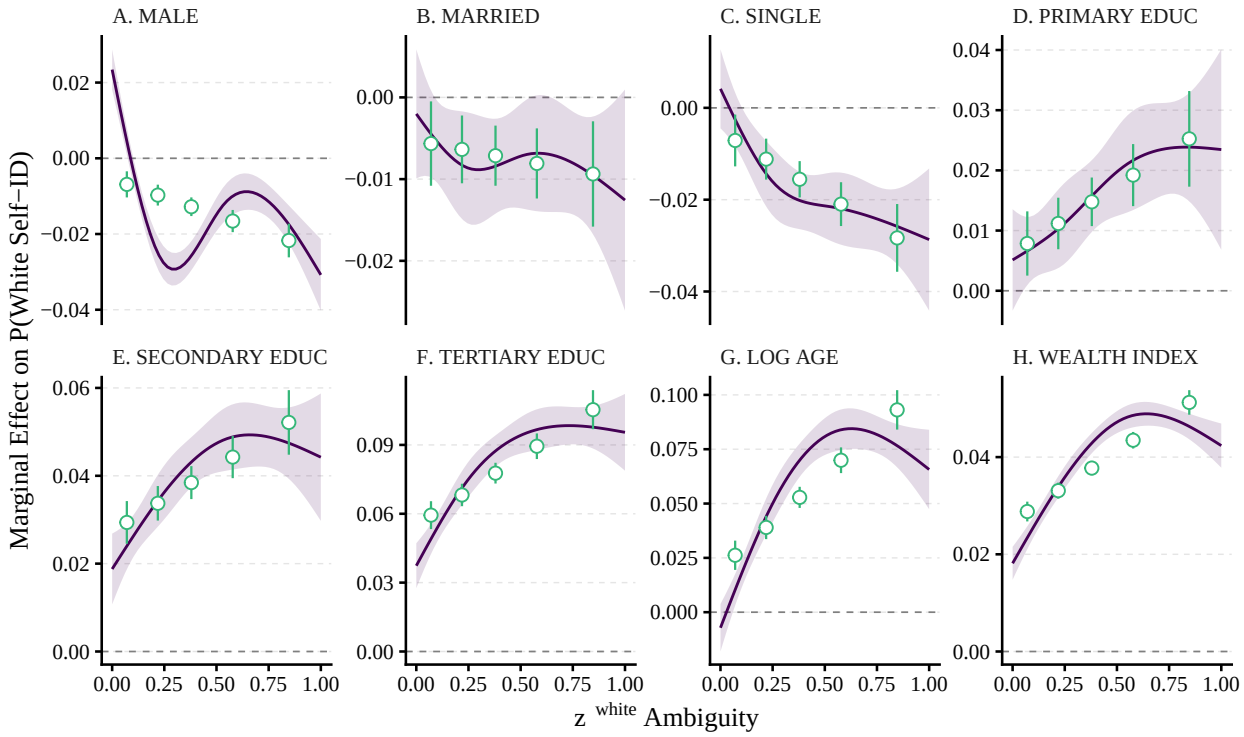

Figure C.2: **Socioeconomic effects by racial ambiguity—white-centered measure.** Marginal effects of socioeconomic variables on the probability of self-identifying as white, as a function of white-centered ambiguity $A_i^{wc} = 1 - 2|z_i^{\text{white}} - 0.5|$. The curves show estimates from a cubic spline interaction model with 95% confidence bands; the points show estimates from a linear interaction model evaluated at quintile midpoints with 95% confidence intervals. Both specifications include appearance controls and municipality fixed effects. Standard errors clustered at the municipality level.

Raw $\widehat{p}^{\text{white}}$. Figure C.3 replaces the ambiguity measure with the raw logit-based probability $\widehat{p}_i^{\text{white}}$, so the x-axis runs from unambiguously nonwhite (left) to unambiguously white (right). This specification is informative because it breaks the symmetry imposed by the ambiguity transformation, separating candidates at opposite ends of the appearance distribution.

The previously monotone gradients become concave, with effects peaking among racially ambiguous candidates (intermediate $\widehat{p}^{\text{white}}$) and declining toward both extremes. Tertiary education increases from 6.1 pp at the lowest quintile ($\widehat{p}^{\text{white}} \approx 0.13$) to 7.8 pp in the middle (≈ 0.46), reaching 9.6 pp at the highest (≈ 0.80). This non-monotonicity reveals that the two groups pooled together at low ambiguity in the main specification are not symmetric: education's effect on white self-identification is substantially larger among candidates who appear unambiguously white than

among those who appear unambiguously nonwhite, suggesting that education reinforces white identity most powerfully when appearance already supports it. Log age shows a similar but flatter pattern, rising modestly from 4.7 to 6.9 pp. The wealth index is nearly flat across the distribution (3.9 to 4.3 pp), indicating that wealth operates similarly regardless of appearance. The male coefficient decreases monotonically from positive (+1.2 pp at low $\widehat{p}^{\text{white}}$) to negative (−1.9 pp at high $\widehat{p}^{\text{white}}$), confirming that the gender effect reversal is robust.

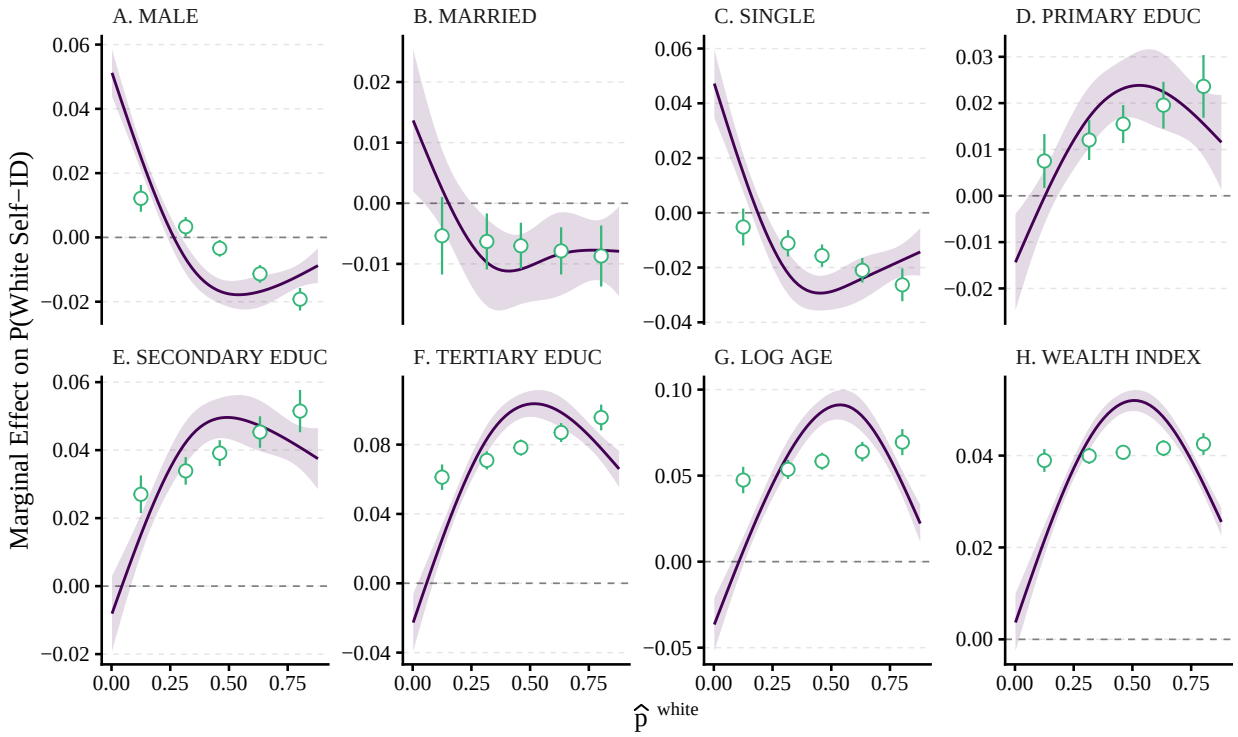

Figure C.3: **Socioeconomic effects by $\widehat{p}^{\text{white}}$.** Marginal effects of socioeconomic variables on the probability of self-identifying as white, as a function of the raw logit-based probability $\widehat{p}_i^{\text{white}}$. Low values indicate individuals whose appearance is unambiguously nonwhite; high values indicate unambiguously white. The curves show estimates from a cubic spline interaction model with 95% confidence bands; the points show estimates from a linear interaction model evaluated at quintile midpoints with 95% confidence intervals. Both specifications include appearance controls and municipality fixed effects. Standard errors clustered at the municipality level.

Appendix D Extensions: Slavery

This appendix examines the robustness of the slavery heterogeneity analysis presented in Figure 10. In the main text, the marginal effect of the 1872 slave population share on white self-identification is plotted against the logit-based ambiguity measure $A_i = 1 - 2|\widehat{p}_i^{\text{white}} - 0.5|$ for the 2020 cross-section, where the linear interaction term is 0.657 ($p < 0.01$) and the marginal effect rises from 0.100 at the lowest quintile to 0.593 at the highest. Figure D.1 replicates this analysis using the white-centered ambiguity measure (Panel A) and the raw logit-based probability $\widehat{p}_i^{\text{white}}$ (Panel B). All specifications match column (2) of Table 4: state fixed effects, appearance controls, geography controls, and a second-order polynomial of latitude and longitude, with standard errors clustered at the 1872 AMC level.

Panel A uses the white-centered ambiguity measure $A_i^{wc} = 1 - 2|z_i^{\text{white}} - 0.5|$. The linear interaction term is 0.586 ($p < 0.01$), and the marginal effect rises from 0.157 at the lowest ambiguity quintile ($A^{wc} \approx 0.07$) to 0.612 at the highest ($A^{wc} \approx 0.85$). The pattern closely mirrors the main text: the effect is near zero and statistically insignificant for individuals with unambiguous appearance, and increases monotonically with ambiguity. The spline specification confirms a smooth, roughly linear gradient—all three spline–slavery interaction terms are jointly significant at the 1% level. The similarity between the two ambiguity measures confirms that the key finding is not specific to the logit-based construction of A_i.

Panel B replaces the ambiguity transformation with the raw logit-based probability $\widehat{p}_i^{\text{white}}$, so that the x-axis runs from unambiguously nonwhite (left) to unambiguously white (right). The linear interaction term is 0.832 ($p < 0.01$), and the marginal effect increases from 0.063 (insignificant) at $\widehat{p}^{\text{white}} \approx 0.13$ to 0.627 at $\widehat{p}^{\text{white}} \approx 0.80$. Unlike the ambiguity measures, which impose symmetry around 0.5, the raw probability reveals the full asymmetry: the slavery association is negligible among clearly nonwhite individuals and grows monotonically with $\widehat{p}^{\text{white}}$. However, the spline specification reveals that the marginal effect levels off—and begins to decline—at the highest values of $\widehat{p}^{\text{white}}$, tracing a concave shape. This is consistent with the slavery effect operating most strongly near the classification boundary: for clearly white individuals, appearance already

dominates self-identification, leaving less room for historical context to shift the outcome. The inverted-U pattern visible in the spline thus reconciles the monotone gradient in A_i with the expectation that the effect should attenuate at both extremes of the appearance distribution.

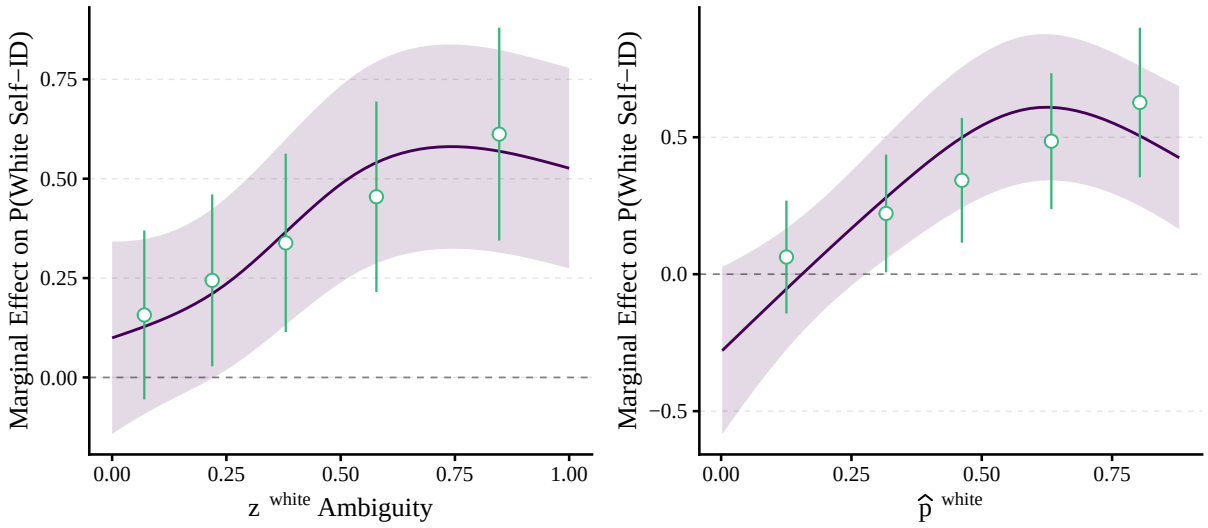

Figure D.1: **Historical slavery and racial self-identification—robustness.** Marginal effect of the 1872 slave population share on the probability of self-identifying as white. Panel A uses the white-centered ambiguity measure $A_i^{wc} = 1 - 2|z_i^{\text{white}} - 0.5|$. Panel B plots against the raw logit-based probability $\widehat{p}_i^{\text{white}}$. The curves show estimates from a cubic spline interaction model with 90% confidence bands; the points show estimates from a linear interaction model evaluated at quintile midpoints with 90% confidence intervals. All specifications include state fixed effects, appearance controls, geography controls, and a second-order polynomial of latitude and longitude. Standard errors clustered at the 1872 AMC level.

Appendix E Extensions: Wealth Sample

Candidates in Brazilian elections are required to disclose the value of their assets, which we use as a proxy for wealth. This measure has important limitations. There is no formal verification of the reported values, although physical assets such as houses and vehicles may be visible and subject to public scrutiny. In contrast, bank accounts are not publicly observable and are more likely to be omitted. As a result, many candidates report zero wealth. Souto-Maior and Borba (2019) suggests that some candidates may believe only high-value assets must be declared or that those exempt from income taxes are not required to report, leading to underreporting. We replicate our main analyses using only individuals who reported positive wealth (approximately 322,000 in the 2020 cross-section).

Shapley decomposition. Figure E.1 presents the Shapley-Owen decomposition for the wealth sample. Panel A shows the aggregate decomposition: appearance accounts for 0.164 (40.2%), socioeconomic characteristics for 0.023 (5.5%), and municipality fixed effects for 0.222 (54.3%), for a total adjusted R^2 of 0.409. These shares are closely aligned with the full-sample decomposition (0.166, 0.021, and 0.209 for a total of 0.396), confirming that restricting to candidates with positive wealth does not alter the relative importance of the three groups.

Panel B replicates the ambiguity heterogeneity analysis within the wealth sample. The pattern mirrors the main text: appearance contributions decline from 0.474 (66.3%) in the lowest ambiguity quintile to 0.011 (3.8%) in the highest, while municipality fixed effects rise from 28.6% to 89.1% of explained variation. Total R^2 falls from 0.72 to 0.28 across quintiles. These gradients are virtually identical to those in the full sample (0.475 to 0.011 for appearance; 27.8% to 89.4% for municipality fixed effects), indicating that the decomposition results are robust to sample composition.

Socioeconomic heterogeneity. Figure E.2 presents the socioeconomic heterogeneity analysis for the wealth sample, where log wealth enters as an additional control alongside education, age, gender, marital status, and occupation. The education gradients are preserved: tertiary education

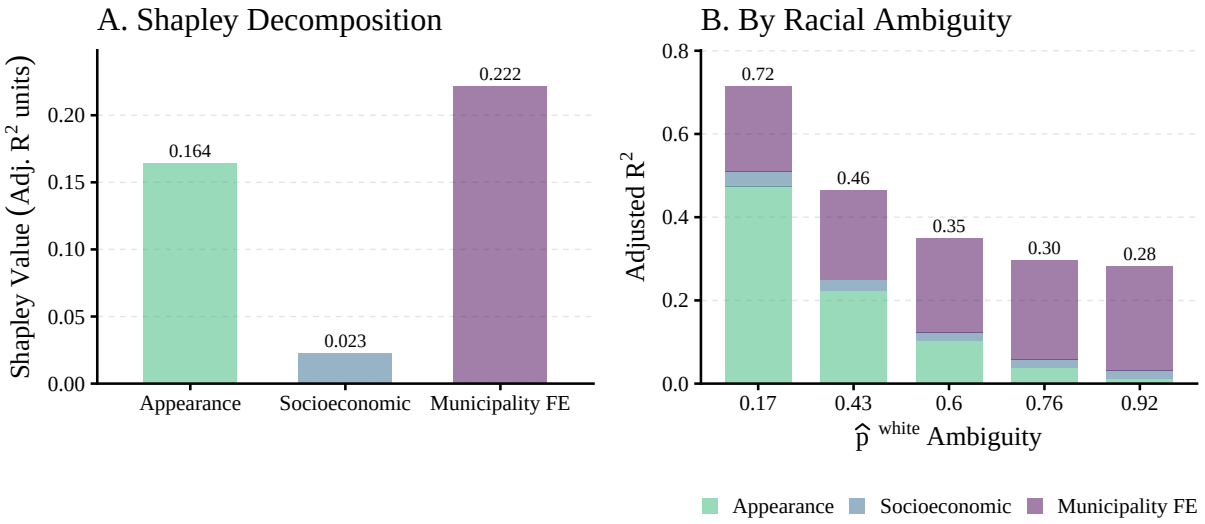

Figure E.1: **Shapley-Owen decomposition—wealth sample.** Panel A shows Shapley values in adjusted R^2 units for three groups of regressors in the wealth sample (candidates with positive declared assets). Panel B repeats the decomposition within quintiles of logit-based racial ambiguity $A_i = 1 - 2|\widehat{p}_i^{\text{white}} - 0.5|$. Each stacked bar's total height equals the full-model adjusted R^2 for that quintile; colored segments show each component's contribution.

increases white self-identification by 3.2 percentage points at low ambiguity ($A \approx 0.17$), rising to 9.4 percentage points at high ambiguity ($A \approx 0.92$), compared to 3.6–11.3 percentage points in the full-sample main specification. Secondary education follows a similar pattern (1.6 to 4.4 pp). Log age displays a steep gradient, rising from near zero (−0.2 pp) to 9.2 pp, consistent with the main specification.

Log wealth itself exhibits an increasing gradient with ambiguity: a one log-point increase in reported wealth is associated with a 0.6 percentage point increase in white self-identification at low ambiguity, rising to 1.6 percentage points at high ambiguity. This confirms that the social whitening pattern documented for the wealth index in the main specification also holds for directly observed wealth, and that the effect is concentrated among racially ambiguous individuals. The inclusion of log wealth does not substantially alter the education coefficients, consistent with education and wealth operating through partially distinct channels.

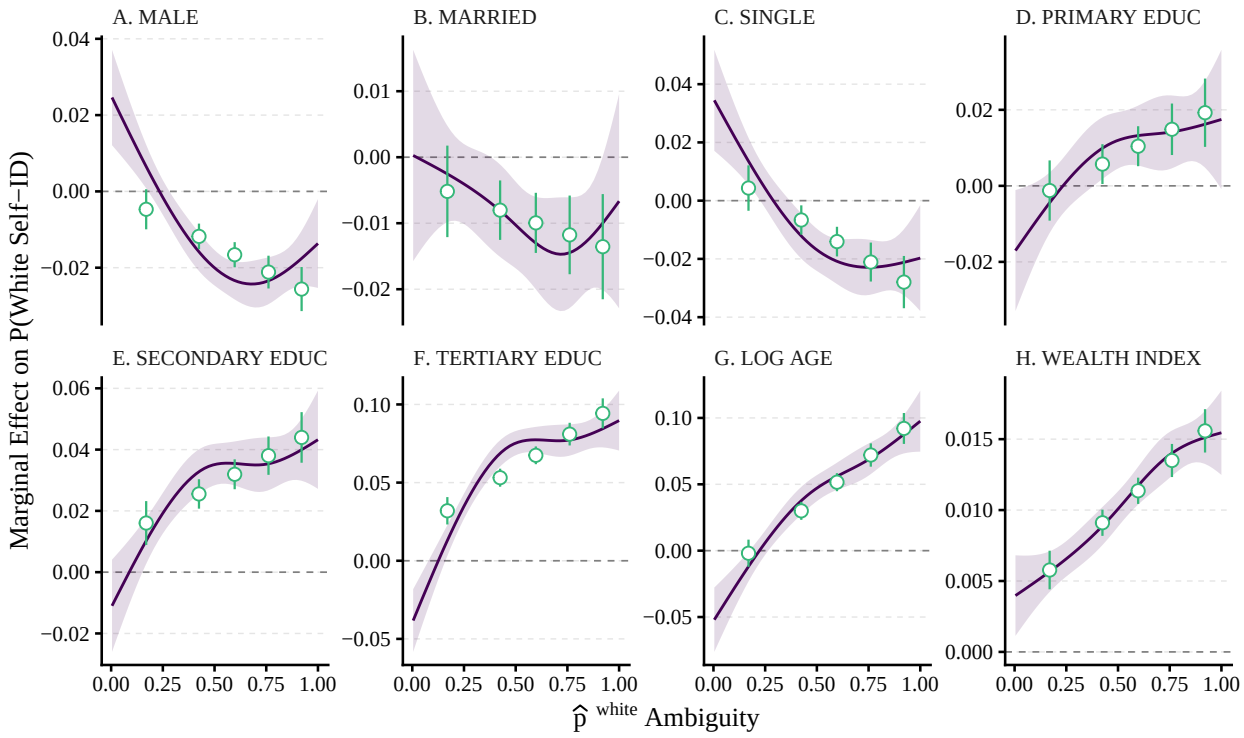

Figure E.2: **Socioeconomic effects by racial ambiguity—wealth sample.** Marginal effects of socioeconomic variables on the probability of self-identifying as white, as a function of logit-based racial ambiguity $A_i = 1 - 2|\widehat{p}_i^{\text{white}} - 0.5|$. Sample restricted to candidates with positive declared assets; log wealth is included as an additional control. The curves show estimates from a cubic spline interaction model with 95% confidence bands; the points show estimates from a linear interaction model evaluated at quintile midpoints with 95% confidence intervals. Both specifications include appearance controls and municipality fixed effects. Standard errors clustered at the municipality level.

References

Adukia, Anjali, Richard Hornbeck, Daniel Keniston, and Benjamin Lualdi, "The Social Construction of Race during Reconstruction," Working Paper 33502, National Bureau of Economic Research 2025.

Agadjanian, Alexander and Dean Lacy, "Changing votes, changing identities? Racial fluidity and vote switching in the 2012–2016 US Presidential Elections," *Public Opinion Quarterly*, 2021, *85* (3), 737–752.

Akerlof, George A and Rachel E Kranton, *Identity Economics: How Our Identities Shape Our Work, Wages, and Well-Being*, Princeton, NJ: Princeton University Press, 2010.

Alekhin, Alexander, Alexander Smorkalov, and Vadim Pisarevsky, "OpenCV," https://github.com/opencv/opencv 2026.

Andrews, George Reid, "Brazilian Racial Democracy, 1900-90: An American Counterpoint," *Journal of Contemporary History*, 1996, *31* (3), 483–507.

Angrist, Joshua D and Jörn-Steffen Pischke, *Mostly Harmless Econometrics: An Empiricist's Companion*, Princeton, NJ: Princeton University Press, 2009.

Antman, Francisca M and Brian Duncan, "American Indian Casinos and Native American Self-Identification," *Journal of the European Economic Association*, 2023, *21* (6), 2547–2585.

__ and __, "Ethnic Identity and Anti-Immigrant Sentiment: Evidence from Proposition 187," Technical Report, National Bureau of Economic Research 2024.

Baerlocher, Diogo and Rodrigo Schneider, "Racial Self-Classification, Group Consciousness, and Public Employment Representation," *Journal of Development Economics*, 2026, *181*, 103755.

Bailey, Stanley R and Edward E Telles, "Multiracial versus collective black categories: examining census classification debates in Brazil," *Ethnicities*, 2006, *6* (1), 74–101.

Barsanetti, Bruno and A. Ferreira, "Georeferenced and Cataloged Dataset Built from Nimuendajú's Ethnohistorical Map," 2024. Version 2, July 2024. Data set.

Bernheim, B. Douglas, "A Theory of Conformity," *Journal of Political Economy*, 1994, *102* (5), 841–877.

Camargo, Alexandre de Paiva Rio, "Mensuração racial e campo estatístico nos censos brasileiros (1872–1940): uma abordagem convergente," *Boletim do Museu Paraense Emílio Goeldi. Ciências Humanas*, 2009, *4* (3).

Chor, Dóra, Alexandre Pereira, Antonio G. Pacheco, Ricardo V. Santos, Maria J. M. Fonseca, Maria I. Schmidt, Bruce B. Duncan, Sandhi M. Barreto, Estela M. L. Aquino, José G. Mill, Maria del C. B. Molina, Luana Giatti, Maria da C. C. Almeida, Isabela Bensenor, and Paulo A. Lotufo, "Context-Dependence of Race Self-Classification: Results from a Highly Mixed and Unequal Middle-Income Country," *PLOS ONE*, 2019, *14* (5), e0216653.

Dahis, Ricardo, Emily Nix, and Nancy Qian, "Choosing racial identity in the United States, 1880-1940," Technical Report, National Bureau of Economic Research 2020.

Deng, Jiankang, Jia Guo, Yuxiang Zhou, Jinke Yu, Irene Kotsia, and Stefanos Zafeiriou, "RetinaFace: Single-stage Dense Face Localisation in the Wild," 2019. Available at https://arxiv.org/abs/1905.00641.

Derenoncourt, Ellora, Chi Hyun Kim, Moritz Kuhn, and Moritz Schularick, "Wealth of two nations: The US racial wealth gap, 1860–2020," *The Quarterly Journal of Economics*, 2024, *139* (2), 693–750.

Dupree-Wilson, Teisha, "Phenotypic proximity: Colorism and intraracial discrimination among Blacks in the United States and Brazil, 1928 to 1988," *Journal of Black Studies*, 2021, *52* (5), 528–546.

FAO/IIASA, "Global Agro-Ecological Zones (GAEZ v3.0)," Technical Report, Food and Agriculture Organization of the United Nations and International Institute for Applied Systems Analysis, Rome and Laxenburg 2012.

Feliciano, Cynthia, "Shades of race: How phenotype and observer characteristics shape racial classification," *American Behavioral Scientist*, 2016, *60* (4), 390–419.

Fournier-Montgieux, Alexandre, Hervé Le Borgne, Adrian Popescu, and Bertrand Luvison, "Reliable and Reproducible Demographic Inference for Fairness in Face Analysis," *Working Paper*, 2025. Available at https://arxiv.org/pdf/2510.20482.

Francis, Andrew M and Maria Tannuri-Pianto, "Endogenous race in Brazil: affirmative action and the construction of racial identity among young adults," *Economic Development and cultural change*, 2013, *61* (4), 731–753.

Francis-Tan, Andrew and Maria Tannuri-Pianto, "Affirmative action in Brazil: global lessons on racial justice and the fight to reduce social inequality," *Oxford Review of Economic Policy*, 2024, *40* (3), 642–655.

Freeman, Nicholas C., Edward E. Telles, and Rachel E. Goldberg, "The Changing Relationship between Racial Identity and Skin Color in Brazil," *Proceedings of the National Academy of Sciences*, 2025, *122* (1), e2411495121.

Htun, Mala, "From "racial democracy" to affirmative action: changing state policy on race in Brazil," *Latin American Research Review*, 2004, *39* (1), 60–89.

Huang, Gary B., Marwan Mattar, Tamara Berg, and Eric Learned-Miller., "Labeled Faces in the Wild: A Database forStudying Face Recognition in Unconstrained Environments," in Erik Learned-Miller, Andras Ferencz, and Frédéric Jurie, eds., *Workshop on Faces in 'Real-Life' Images: Detection, Alignment, and Recognition*, 2008.

IBGE, "Sintese de Indicadores Sociais: Uma análise da condições de vida da população brasileira," Technical Report, Instituto Brasileiro de Geografia e Estatística - IBGE 2024.

Israeli, Osnat, "A Shapley-based decomposition of the R-square of a linear regression," *Journal of Economic Inequality*, 2007, *5* (2), 199–212.

Janusz, Andrew, "Electoral Incentives and Elite Racial Identification: Why Brazilian Politicians Change Their Race," *Electoral Studies*, 2021, *72*, 102340.

Karkkainen, Kimmo and Jungseock Joo, "Fairface: Face attribute dataset for balanced race, gender, and age for bias measurement and mitigation," in "Proceedings of the IEEE/CVF winter conference on applications of computer vision" 2021, pp. 1548–1558.

Kumar, Neeraj, Alexander C. Berg, Peter N. Belhumeur, and Shree K. Nayar, "Attribute and simile classifiers for face verification," in "2009 IEEE 12th International Conference on Computer Vision" 2009, pp. 365–372.

Lancaster, Tony, "The incidental parameter problem since 1948," *Journal of Econometrics*, 2000, *95* (2), 391–413.

Lang, Kevin and Ariella Kahn-Lang Spitzer, "Race discrimination: An economic perspective," *Journal of Economic Perspectives*, 2020, *34* (2), 68–89.

Laudares, Humberto and Felipe Valencia Caicedo, "Tordesillas, Slavery and the Origins of Brazilian Inequality," Technical Report, CEPR Discussion Papers 2023.

McGee, Dan, "Exploitation through Racialization," *The Quarterly Journal of Economics*, 2025, *140* (2), 1581–1631.

Mello, Ursula, "Centralized admissions, affirmative action, and access of low-income students to higher education," *American Economic Journal: Economic Policy*, 2022, *14* (3), 166–197.

Mitchell, Gladys, "The politics of skin color in Brazil," *The Review of Black Political Economy*, 2010, *37* (1), 25–41.

Monk Jr., Ellis P., "The Consequences of "Race and Color" in Brazil," *Social Problems*, 2016, *63* (3), 413–430.

Morning, Ann, *The Nature of Race: How Scientists Think and Teach about Human Difference*, Berkeley: University of California Press, 2011.

Naritomi, Joana, Rodrigo R Soares, and Juliano J Assunção, "Institutional development and colonial heritage within Brazil," *The journal of economic history*, 2012, *72* (2), 393–422.

Nimuendajú, Curt, *Mapa Etno-Histórico do Brasil e Regiões Adjacentes*, 2nd ed., Rio de Janeiro: IPHAN/IBGE, 2017.

Nunn, Nathan and Leonard Wantchekon, "The slave trade and the origins of mistrust in Africa," *American economic review*, 2011, *101* (7), 3221–3252.

Omi, Michael and Howard Winant, *Racial Formation in the United States*, 3rd ed., New York: Routledge, 2014.

Parkhi, Omkar M., Andrea Vedaldi, and Andrew Zisserman, "Deep Face Recognition," in "British Machine Vision Conference" 2015.

Pereira, RHM, Caio N Gonçalves et al., "geobr: loads shapefiles of official spatial data sets of Brazil," *GitHub repository*, 2019.

Portella, Alysson, Charles Kirschbaum, and Naercio Menezes-Filho, "Racial Social Norms among Brazilian Students: Academic Performance, Popularity, and Racial Identification," *Proceedings of the National Academy of Sciences*, 2022, *119* (27), e2117956119.

Reis, Eustáquio, Márcia Pimentel, Ana I Alvarenga, and Maria do Carmo H Santos, "Áreas mínimas comparáveis para os períodos intercensitários de 1872 a 2000," in "Anais do I Simpósio Brasileiro de Cartografia Histórica" Paraty 2011.

Rocha, Rudi, Claudio Ferraz, and Rodrigo R Soares, "Human capital persistence and development," *American Economic Journal: Applied Economics*, 2017, *9* (4), 105–136.

Santos, Sales Augusto Dos, "Historical roots of the "Whitening" of Brazil," *Latin American Perspectives*, 2002, *29* (1), 61–82.

Saperstein, Aliya, "Recognizing Identity Fluidity in Demographic Research," *Population and Development Review*, 2025, *51* (1), 519–538.

_ **and Andrew M. Penner**, "Racial Fluidity and Inequality in the United States," *American Journal of Sociology*, 2012, *118* (3), 676–727.

Schwartzman, Luisa Farah, "Does Money Whiten? Intergenerational Changes in Racial Classification in Brazil," *American Sociological Review*, 2007, *72* (6), 940–963.

Serengil, Sefik, "DeepFace," https://github.com/serengil/deepface 2026.

_ **and Alper Ozpinar**, "A Benchmark of Facial Recognition Pipelines and Co-Usability Performances of Modules," *Journal of Information Technologies*, 2024, *17* (2), 95–107.

Shorrocks, Anthony F., "Decomposition procedures for distributional analysis: a unified framework based on the Shapley value," *Journal of Economic Inequality*, 2013, *11* (1), 99–126.

Skidmore, Thomas E, *Brazil: Five centuries of change*, Oxford University Press, 2009.

Souto-Maior, Cesar Duarte and José Alonso Borba, "Consistência na declaração de bens dos candidatos nas eleições brasileiras: ficção ou realidade?," *Revista de Administração Pública*, 2019, *53*, 195–213.

Telles, Edward and Tianna Paschel, "Who is black, white, or mixed race? How skin color, status, and nation shape racial classification in Latin America," *American Journal of Sociology*, 2014, *120* (3), 864–907.

Telles, Edward E, *Race in another America: The significance of skin color in Brazil*, Princeton University Press, 2004.

Vieira, Renato Schwambach and Mary Arends-Kuenning, "Affirmative action in Brazilian universities: Effects on the enrollment of targeted groups," *Economics of Education Review*, 2019, *73*, 101931.